

On generalizations of the Lai–Massey scheme

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Abstract. In this paper, we re-investigate the Lai-Massey scheme, originally proposed in the cipher IDEA. Due to the similarity with the Feistel networks, and due to the existence of invariant subspace attacks as originally pointed out by Vaudenay at FSE 1999, the Lai-Massey scheme has received only little attention by the community. As first contribution, we propose two new generalizations of such scheme that are not (extended) affine equivalent to any generalized Feistel network proposed in the literature so far. Then, inspired by the recent **Horst** construction, we propose the **Amaryllises** structure as a generalization of the Lai-Massey scheme, in which the linear combination in the Lai-Massey scheme can be replaced by a non-linear one. Besides proposing concrete examples of the **Amaryllises** construction, we analyze its cryptographic properties in the context of MPC-/HE-/ZK-friendly symmetric primitives.

Keywords: Lai-Massey · Generalized/Redundant Lai-Massey · Amaryllises · Generalized Feistel · Horst

1 Introduction

Probably, the two most popular design frameworks for iterated symmetric primitives are the Substitution-Permutation Network (SPN) and the Feistel one (FN). In the SPN case, the input of each round is divided into multiple small sub-blocks, and a non-linear function (called S-Box) is applied on each sub-block, followed by an affine transformation that mixes the sub-blocks. The invertibility of the entire construction depends on the invertibility of each sub-component. The scenario is different in the FN case. In each round of a Feistel network, the input is split into two halves, a function $\mathcal{F}$ is applied on one of the two halves, which is successively mixed with the other part, just before the two halves are swapped, that is, $[x_0, x_1] \mapsto [y_0, y_1] := [x_1 + \mathcal{F}(x_0), x_0]$. With respect to the SPN case, FNs are invertible by construction independently of the details of the $\mathcal{F}$ -function. Hence, the designer can choose among a larger class of non-linear functions in order to instantiate a FN with respect to what happens in SPNs. Moreover, the costs of computing a Feistel network in the forward and in the backward direction are very similar (even identical in some cases), since the same $\mathcal{F}$ -function is computed in the two processes. Due to these facts, a large proportion of symmetric primitives adopts the Feistel design approach, including DES, Blowfish [Sch93], MISTY [Mat97], among others, and several generalizations have been proposed in the literature, including the Type-I/-II/-III Feistel networks [ZMI90, Nyb96].

Another design strategy that has many points in common with the FNs is the Lai-Massey scheme [Vau99], introduced after the design of IDEA [LM90]. As in the case of a FN, the input is first split into two halves, but in this case a function $\mathcal{F}$ is applied on their difference, and the result of such function is then added to each input, that is,

$$[x_0, x_1] \mapsto [y_0, y_1] := [x_0 + \mathcal{F}(x_0 - x_1), x_1 + \mathcal{F}(x_0 - x_1)]. \quad (1)$$

Analogous to the FNs, the invertibility of Lai-Massey schemes follows from its construction, that is, it is independent of details of the function $\mathcal{F}$. However, compared to the FNs, the Lai-Massey scheme is much less studied in the literature, and only few concrete Lai-Massey schemes have been proposed in the literature. This is due to several factors, including the following:

1. Lai-Massey schemes as the one just proposed can be easily broken due to the existence of an invariant subspace attack, as first pointed out by Vaudenay [Vau99];
2. Lai-Massey schemes do not (seem to) have any concrete advantage with respect to Feistel networks, as stated by Yun et al. in [YPL11, Sect. 8]: “*as a cryptographic design, the Lai-Massey cipher does not have any advantage over the Feistel in terms of the Luby-Rackoff model*”.

In this paper, we re-consider the Lai-Massey construction, and we present new generalizations of it that are not affine equivalent to any generalized Feistel network proposed in the literature so far. Moreover, we introduce the generalized **Amaryllises** construction, a new generalization of the Lai-Massey one in which the linear combination between the function $\mathcal{F}$ and the halves that composed the input can be replaced by a non-linear combination.

1.1 Generalized and Redundant Lai-Massey Schemes

Relation between Generalized Feistel and Lai-Massey Schemes The simplest generalization of a Lai-Massey scheme recently proposed in [GØSW23] and recalled in Sect. 3 works as follows: (1st) the input message is divided in $n \geq 2$ sub-blocks, then (2nd) a function $\mathcal{F}$ is applied to zero-sum linear combinations of such sub-blocks (that is, linear combinations whose coefficients sum to zero), and finally (3rd) the result of such function is then added to each input.

In Sect. 3.2, we prove that any Lai-Massey scheme of that form is *affine equivalent* to a generalized Feistel network, that is, a Lai-Massey scheme of this form is equal to a generalized FN pre- and post-processed with affine invertible transformations. Moreover, we show that any iterated symmetric primitive instantiated with a Lai-Massey scheme is affine equivalent to an iterated scheme whose round function is a Feistel network *without* the rotation.

Remark 1. For completeness, we point out that the result just proposed is not completely new in the literature. For example, in [YPL11], Yun et al. introduced the concept of “quasi-Feistel” network, a generic class of primitives over finite quasi-groups that includes as special cases both the Feistel networks and the Lai-Massey schemes. With respect to the results published in the literature such as [YPL11], our result points out the relation between Feistel networks and Lai-Massey schemes by directly showing the affine equivalence, without introducing any new function/construction.

New Generalizations of the Lai-Massey Schemes Based on the previous results, it seems there is no concrete reason to prefer a Lai-Massey scheme with respect to a Feistel network. Indeed, at round level, the two primitives have the same statistical (linear and differential) properties.¹ Still, it is possible that generalizations of the Lai-Massey schemes exist such that (i) they are not affine equivalent to any Feistel network (hence, with potentially better linear and differential properties), but (ii) they still preserve the “properties” that characterize a Lai-Massey scheme.

For this reason, as next step, in Sect. 4 and 5, we propose two new generalizations of the Lai-Massey scheme which aim to capture the “essence” of a Lai-Massey scheme. They are:

- the **generalized Lai-Massey schemes**: instead of adding the same fixed function to each input, we allow for *different functions*. However, we still impose that such functions take zero-sum linear combinations of the sub-blocks, and that the entire scheme is invertible;
- the **redundant Lai-Massey schemes**: instead of limiting ourselves to consider a function which takes as inputs zero-sum linear combinations of the sub-blocks, we allow for *any fixed function* $\mathcal{F}$ for which the entire construction is invertible.

Formal definitions are given in Sect. 4 and 5, respectively Def. 5 and Def. 6, together with concrete examples of generalized and redundant Lai-Massey schemes over a field $\mathbb{F}_q^n$ (where $q = p^s$ for a prime $p \geq 2$ and a positive integer s) that are *not* extended affine equivalent to any generalized Feistel network.

1.2 Amaryllises for MPC-/HE-/ZK-Friendly Symmetric Schemes

MPC-/HE-/ZK-Friendly Symmetric Primitives Currently, one of the hottest topics in symmetric cryptography regards the design and the analysis of symmetric primitives for applications such as Multi-Party Computation (MPC), Homomorphic Encryption (HE), and Zero-Knowledge (ZK). Those applications require dedicated symmetric primitives that *minimize the number of field multiplications required to compute and/or verify the primitive in their natural algorithmic descriptions*. In contrast to traditional/classical symmetric primitives like AES and Keccak/SHA-3 defined over $\mathbb{F}_{2^n}^t$ for small

¹ We recall that the situation is different when considering an iterated scheme instantiated with a Lai-Massey scheme or a Feistel network, and an additional linear layer. In such a case, the details of the linear layer play a crucial role on the security argument over multiple rounds.

$n \in \{3, 4, \dots, 8\}$, these new MPC-/HE-/ZK-friendly primitives usually operate over a vector space $\mathbb{F}_p^t$ for a large prime p such as $p \approx 2^{128}$ or $p \approx 2^{256}$.

The particular cost metric these MPC-/HE-/ZK-friendly symmetric primitives aim to minimize has a crucial impact on their design strategy. Due to the large size of p , no function can be pre-computed and stored as a look-up table. Hence, a MPC-/HE-/ZK-friendly symmetric primitive must admit a *simple algebraic expression*. For example, the majority of the MPC-/HE-/ZK-friendly symmetric primitives are instantiated with invertible power maps $x \mapsto x^d$ over $\mathbb{F}_p$. Besides, this particular design approach usually allows also to minimize the *multiplicative complexity*, that is, the number of multiplications for evaluating and/or verifying the system of polynomials equations associated to the symmetric primitive. As a direct consequence of this fact, algebraic attacks are usually much stronger than statistical attacks in the case of MPC-/HE-/ZK-friendly primitives, and several MPC-/HE-/ZK-friendly symmetric primitives have been indeed broken via algebraic attacks [ACG⁺19, BCD⁺20, BBLP22, GKRS22, BBLP22, Bar23, KLR24, BBL⁺24].

Due to all these facts, the different cost metrics these primitives aim to minimize pushed the designers to look for *new design strategies*. Indeed, the traditional and most common design strategies exploited in symmetric cryptography are not necessarily well suited for MPC-/HE-/ZK-friendly schemes in general. Without the goal of being exhaustive, examples of some recent innovative and dedicated design strategies include:

- **Horst in Griffin**: the **Horst** construction recently introduced by Grassi et al. [GHR⁺23] is a variant of the Feistel network, in which a non-linear mixing takes place. Over $\mathbb{F}_q^2$, it is defined as $[x_0, x_1] \mapsto [y_0, y_1] := [x_1 \cdot \mathcal{G}(x_0) + \mathcal{F}(x_0), x_0]$. Even if a single round of **Horst** is in general more expensive than a single round of a Feistel network, the **Griffin**'s designers showed that such construction has concrete advantages in order to defeat the Gröbner basis attack over multiple rounds compared to a Feistel network;
- **Flystel in Anemoi**: the **Flystel** is a particular 3-round Feistel network over $\mathbb{F}_q^2$ introduced by Bouvier et al. in [BBC⁺23]. Over a prime field, its rounds are instantiated via the power maps $x \mapsto x^2$ and $x \mapsto x^{1/d}$, where $d \geq 3$ is the smallest integer co-prime with $p - 1$. The advantage of such non-linear function relies on the cheap cost necessary to verify it. For this reason, it is used to instantiate the S-Boxes of the SPN scheme **ANEMOI**;
- **Generalized Triangular Dynamical System (GTDS) in Arion**: the Generalized Triangular Dynamical System [RS22] is a non-linear layer defined via the combination/concatenation of a SPN's S-Box layer with a **Horst** construction. It instantiates the non-linear of the hash function **Arion** [RST23].

The Blooming of Amaryllises The details of the non-linear layer of a MPC-/HE-/ZK-friendly primitive play a crucial role for what concerning the security and the performance of the primitive itself. Indeed, remember that such primitives aim to minimize the multiplicative complexity, and at the same time that they are particularly vulnerable to algebraic attacks. Based on this, it is crucial to design new and innovative dedicated non-linear layers that can be used to design new MPC-/HE-/ZK-friendly primitive in such a way to improve their efficiency without affecting the security.

We concretely face this problem in this paper. Taking inspiration from the **Horst** construction, we propose a variant of the Lai-Massey scheme in which the linear combinations can be replaced with non-invertible ones. We call this new construction as “**Amaryllises**”.² A formal definition is given in Theorem 2 – see Sect. 6. In there, we also show how to concretely instantiate it in an efficient way.

Next, in Sect. 7, we propose an initial generic analysis of its statistical and algebraic properties, and we discuss some analogies and possible advantages with respect to other non-linear constructions proposed in the literature so far in the context of MPC-/HE-/ZK-friendly schemes. Similar to MPC-/HE-/ZK-friendly SPN schemes, the degree in a **Amaryllises** can grow much faster in the backward direction than in the forward direction, a crucial fact for preventing Meet-in-the-Middle algebraic attacks in an efficient way. At the same time, with respect to SPN schemes, full diffusion takes place in **Amaryllises** in every round, without the needed of extra linear layer.

Finally, we present a possible generalization of **Amaryllises** called the Contracting-**Amaryllises** construction in Sect. 8.

² We decided to call it as the flowers *ama(r)yl(l)ises*, which is (almost) the anagram of Lai-Massey.

2 Preliminary

Notation. Let $q = p^s$ where $p \geq 2$ is a prime number and $s \geq 1$ is a positive integer. Let $\mathbb{F}_q$ denote the Galois Field of order q . We use small letters to denote both indexes and variables, and greek letters to denote fixed elements (as parameters) in $\mathbb{F}_q$. We use the calligraphic font (e.g., $\mathcal{F}$) to denote functions, with the only exceptions of linear/affine functions denoted via the capital font. We use the frankfurt font (e.g., $\mathfrak{X}$) to denote sets of elements, and $|\cdot|$ to denote their cardinality. Given $x \in \mathbb{F}_q^n$, we denote by x_i its i -th component for each $i \in \{0, 1, \dots, n-1\}$, that is, $x = [x_0, x_1, \dots, x_{n-1}] \equiv x_0 \| x_1 \| \dots \| x_{n-1}$, where the symbol $\cdot \| \cdot$ denotes concatenation. Given a matrix $M \in \mathbb{F}_q^{n \times m}$, we denote the entry in the r -th row and in the c -th column by $M_{r,c}$. We use $\langle s^{(0)}, s^{(1)}, \dots, s^{(t-1)} \rangle \subseteq \mathbb{F}_q^n$ to denote the linear span of the vectors $s^{(0)}, s^{(1)}, \dots, s^{(t-1)} \in \mathbb{F}_q^n$.

For the follow-up, we introduce the following definition.

Definition 1. Let $n \geq 2$ be an integer, and let $q = p^s$ be as before. Let $1 \leq l \leq n-1$ (note that $n-1$ is the maximum number of linearly independent $\mathbb{F}_q^n$ -vectors whose entries sum to zero). We say that the sets $\{\lambda_j^{(0)}\}_{j \in \{0,1,\dots,n-1\}}, \dots, \{\lambda_j^{(l-1)}\}_{j \in \{0,1,\dots,n-1\}}$ with $\lambda_j^{(i)} \in \mathbb{F}_q$ for each i, j are “zero-sum linearly independent” if the following conditions are satisfied: (1st) $\sum_{j=0}^{n-1} \lambda_j^{(i)} = 0$ for each $i \in \{0, 1, \dots, l-1\}$, and (2nd) the vectors $[\lambda_0^{(0)}, \lambda_1^{(0)}, \dots, \lambda_{n-1}^{(0)}], [\lambda_0^{(1)}, \lambda_1^{(1)}, \dots, \lambda_{n-1}^{(1)}], \dots, [\lambda_0^{(l-1)}, \lambda_1^{(l-1)}, \dots, \lambda_{n-1}^{(l-1)}]$ are linearly independent.

Generalized Feistel Networks and Horst Constructions. Regarding the definition of generalized Feistel networks, we propose the following:

Definition 2 (Generalized Feistel Schemes). Let $q = p^s$ be as before, and let $n \geq 2$. For each $i \in \{1, 2, \dots, n-1\}$, let $\mathcal{F}_i : \mathbb{F}_q^i \rightarrow \mathbb{F}_q$ be a function. The generalized Feistel network GF over $\mathbb{F}_q^n$ is defined as $\text{GF}(x_0, x_1, \dots, x_{n-1}) := y_0 \| y_1 \| \dots \| y_{n-1}$ where

$$y_i := \begin{cases} x_{i+1} + \mathcal{F}_{i+1}(x_0, x_1, \dots, x_i) & \text{if } i \in \{0, 1, \dots, n-2\} \\ x_0 & \text{otherwise (if } i = n-1) \end{cases}.$$

It is not difficult to check that any Feistel network proposed in the literature including [ZMI90, Nyb96, HR10, BS13] satisfy the definition just given. As it is well known, the invertibility of the entire construction is independent of the details of $\mathcal{F}_0, \dots, \mathcal{F}_{n-2}$. Indeed, we have that (i) $x_0 = y_{n-1}$, and (ii) for each $i \geq 1$, $x_i = y_{i-1} - \mathcal{F}_i(x_0, x_1, \dots, x_{i-1})$ where y_{i-1} and $x_0, x_1, \dots, x_{i-1}$ are given.

The Horst construction recently proposed by Grassi et al. [GHR⁺23] is a generalization of the Feistel networks in which the linear combination is replaced by a non-linear one.

Theorem 1 (Horst [GHR⁺23]). Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. For each $i \in \{1, 2, \dots, n-2\}$, let $\mathcal{F}_i : \mathbb{F}_q^i \rightarrow \mathbb{F}_q$ and $\mathcal{G}_i : \mathbb{F}_q^i \rightarrow \mathbb{F}_q \setminus \{0\}$ (note that $\mathcal{G}_i(x_0, x_1, \dots, x_{i-1}) \neq 0$ for each $x_0, x_1, \dots, x_{i-1} \in \mathbb{F}_q$). The Horst construction H over $\mathbb{F}_q^n$ defined as $\text{H}(x_0, x_1, \dots, x_{n-1}) := y_0 \| y_1 \| \dots \| y_{n-1}$ where

$$y_i := \begin{cases} x_{i+1} \cdot \mathcal{G}_{i+1}(x_0, x_1, \dots, x_i) + \mathcal{F}_{i+1}(x_0, x_1, \dots, x_i) & \text{if } i \in \{0, 1, \dots, n-2\} \\ x_0 & \text{otherwise (if } i = n-1) \end{cases}$$

is invertible.

The invertibility holds due to the same argument previously given for the Feistel networks, due to the fact that $\mathcal{G}$ always returns a non-zero element.

Extended-Affine (EA) Equivalence. Two functions $\mathcal{F}$ and $\mathcal{G}$ are Extended-Affine (EA) equivalent if they satisfy the following requirement.

Definition 3 (EA-Equivalence). Let $q = p^s$ be as before. Let $n, m \geq 1$, and let $\mathcal{F}, \mathcal{G} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ be two functions. $\mathcal{F}$ and $\mathcal{G}$ are extended-affine equivalent (EA-equivalent) if there exist two affine permutations $A : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ and $B : \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m$, and an affine function $C : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ such that

$$\forall x \in \mathbb{F}_q^n, \quad \mathcal{F}(x) = B \circ \mathcal{G} \circ A(x) + C(x).$$

If C is identically equal to zero, then we speak of affine equivalence (A-equivalence).

We limit ourselves to recall that if two functions $\mathcal{F}$ and $\mathcal{G}$ are EA-equivalent, then they share several statistical properties (besides e.g. having the same degree). As a concrete example, the maximum differential probability $\text{DP}_{\max}$ of $\mathcal{F}$ and of $\mathcal{G}$ are equal, where $\text{DP}_{\max}(\mathcal{H}) := \max_{\delta \neq 0, \Delta} |\{x \in \mathbb{F}_q^n \mid \mathcal{H}(x + \delta) - \mathcal{H}(x) = \Delta\}|/q^n$ for a function $\mathcal{H}$ over $\mathbb{F}_q^n$. We refer to [CCZ98, CP19] for more details.

A generalization of the EA-equivalence is the CCZ equivalence defined in [CCZ98, CP19]

Definition 4 (CCZ Equivalence). *Let $q = p^s$ where $p \geq 2$ is a prime and s is a positive integer, and let $n, m \geq 1$. Let $\mathcal{F}, \mathcal{G} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$. The functions $\mathcal{F}$ and $\mathcal{G}$ are CCZ-equivalent if there exists an affine transformation A over $\mathbb{F}_q^n \times \mathbb{F}_q^m$ such that*

$$\{(x, \mathcal{F}(x)) \mid x \in \mathbb{F}_q^n\} = A(\{(x, \mathcal{G}(x)) \mid x \in \mathbb{F}_q^n\}).$$

3 Lai-Massey Schemes in the Literature

Given a function $\mathcal{F}$ over $\mathbb{F}_q$ for $q = p^s$ as before, the Lai-Massey scheme over $\mathbb{F}_q^2$ introduced in [LM90] is defined as in (1), that is, $y_j = x_j + \mathcal{F}(x_0 - x_1)$ for each $j \in \{0, 1\}$. Its invertibility follows from the fact that $y_0 - y_1 = x_0 - x_1$, and so $x_j = y_j - \mathcal{F}(y_0 - y_1)$ for each $j \in \{0, 1\}$.

3.1 Lai-Massey Schemes over $\mathbb{F}_q^{\geq 2}$ from [GØSW23]

A possible generalization of the Lai-Massey scheme over $\mathbb{F}_q^n$ for $n \geq 2$ has been recently proposed in [GØSW23].

Proposition 1 (Prop. 1, [GØSW23]). *Let $n \geq 2$ be an integer, and let $q = p^s$ be as before. Let $l \in \{1, 2, \dots, n-1\}$. Let $\mathcal{F} : \mathbb{F}_q^l \rightarrow \mathbb{F}_q$ be a function. Let $\{\lambda_j^{(i)}\}_{j \in \{0, 1, \dots, n-1\}, i \in \{0, 1, \dots, l-1\}}$ be l “zero-sum linearly independent” sets as in Def. 1.*

The Lai-Massey scheme LM over $\mathbb{F}_q^n$ defined as $\text{LM}(x_0, x_1, \dots, x_{n-1}) := y_0 \| y_1 \| \dots \| y_{n-1}$ where

$$y_i := x_i + \mathcal{F} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right)$$

for each $i \in \{0, 1, \dots, n-1\}$ is invertible.

As for the case of the Lai-Massey scheme over $\mathbb{F}_q^2$, the invertibility holds since

$$\sum_{j=0}^{n-1} \lambda_j^{(i)} \cdot x_j = \sum_{j=0}^{n-1} \lambda_j^{(i)} \cdot y_j + \mathcal{F} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right) \cdot \underbrace{\sum_{j=0}^{n-1} \lambda_j^{(i)}}_{=0} = \sum_{j=0}^{n-1} \lambda_j^{(i)} \cdot y_j$$

for each $i \in \{0, 1, \dots, l-1\}$, which implies

$$\forall i \in \{0, 1, \dots, n-1\}, \quad x_i = y_i - \mathcal{F} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot y_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot y_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot y_j \right).$$

Existence of Invariant Subspace Trails As already pointed out by Vaudenay in [Vau99] for the $\mathbb{F}_q^2$ case, there could exist invariant subspaces for the Lai-Massey scheme proposed in Prop. 1. Following [GRR16],³ we recall that a set $\mathfrak{X}$ is *invariant* for a keyed/unkeyed function $\mathcal{F}_k$ over $\mathbb{F}_q^n$ if for each key/constant $k \in \mathbb{F}_q^n$ and for each $\beta \in \mathbb{F}_q^n$, there exists $\gamma \in \mathbb{F}_q^n$ such that

$$\mathcal{F}_k(\mathfrak{X} + \beta) := \{\mathcal{F}_k(x) \mid x \in \mathfrak{X} + \beta\} = \mathfrak{X} + \gamma. \quad (2)$$

³ For completeness, we point out that we limit ourselves to consider subspaces that are invariant independently of the value of the secret key/constant. In this sense, the definition used here is different from the one proposed in [LAAZ11, LMR15], which require the existence of weak keys/constants for which the subspace is invariant.

As a concrete example, the Lai-Massey scheme LM defined as in Prop. 1 over $\mathbb{F}_q^n$ admits

$$\mathfrak{X} := \left\{ x \in \mathbb{F}_q^n \mid \forall i \in \{0, 1, \dots, l-1\}, \sum_{j=0}^{n-1} \lambda_j^{(i)} \cdot x_j = 0 \right\} \quad (3)$$

as an *invariant* subspace. Note that such set is never empty as

$$\langle [1, 1, \dots, 1] \rangle \equiv \{ [x, x, \dots, x] \in \mathbb{F}_q^n \mid x \in \mathbb{F}_q \} \subseteq \mathfrak{X}$$

independently of the values of $\lambda_j^{(i)}$, since $\sum_{j=0}^{n-1} (\lambda_j^{(i)} \cdot x) = x \cdot (\sum_{j=0}^{n-1} \lambda_j^{(i)}) = 0$ for each $i \in \{0, 1, \dots, l-1\}$, due to the assumption on $\lambda_j^{(i)}$.

More generally, it is not hard to check that $\dim(\mathfrak{X}) = n - \dim(\langle [\lambda_0^{(0)}, \lambda_1^{(0)}, \dots, \lambda_{n-1}^{(0)}], \dots, [\lambda_0^{(l-1)}, \lambda_1^{(l-1)}, \dots, \lambda_{n-1}^{(l-1)}] \rangle) = n - l \geq 1$, due to the linear-independent assumption on the sets $\{\lambda_j^{(i)}\}_j$.

How to Break such Invariant Subspace Trails? Several strategies are possible for destroying such invariant subspace trails. One possibility consists in multiplying the output of LM with an invertible matrix $M \in \mathbb{F}_q^{n \times n}$ that does *not* admit any invariant subspace, that is,

$$\forall \text{ subspace } \mathfrak{Z} \subseteq \mathbb{F}_q^n, \quad M \times \mathfrak{Z} \neq \mathfrak{Z}.$$

As proved in [GRS21], this condition is satisfied if the *minimal polynomial* of M is of maximum degree and irreducible. In particular, by adapting [GRS21, Prop. 13] to the case of the Lai-Massey scheme, we can easily get the following result.

Proposition 2. *Let $q = p^s$ be as before, and let $n, r \geq 2$ be integers. Consider an r -round iterated scheme over $\mathbb{F}_q^n$, whose round function is defined as the concatenation of the Lai-Massey scheme LM over $\mathbb{F}_q^n$ defined as in Prop. 1, a matrix multiplication $M \in \mathbb{F}_q^n$, and the addition with a random round constant $c \in \mathbb{F}_q^n$, that is,*

$$x \mapsto c + M \times \text{LM}(x).$$

If the minimal polynomial of M is of maximum degree and irreducible, and if $r > n - l$ (where l defined as in Prop. 1), then no invariant subspace trail exists for the considered r -round iterated scheme over $\mathbb{F}_q^n$.

The proof follows from the one given in [GRS21, Prop. 13], and it is based on the fact that M does not admit any invariant subspace.

Case: $l = n - 1$. We limit ourselves to mention that it is possible to destroy such invariant subspaces even by imposing a “weaker” condition on the matrix M . Since this topic is out of the scope of this paper, we refer to e.g. [GØSW23] for more details, and we limit ourselves to analyze in more details the case $l = n - 1$ only, that is, the case in which $\mathcal{F}$ depends on *all* the zero-sum linearly independent combinations of $\{x_0, x_1, \dots, x_{n-1}\}$.

In the case $l = n - 1$, the invariant subspace $\mathfrak{X}$ coincides with $\langle [1, 1, \dots, 1] \rangle$. It follows that such subspace is destroyed if $\langle [1, 1, \dots, 1] \rangle$ is not invariant via M . The simplest matrix that satisfies this condition is a diagonal matrix $M = \text{diag}(\mu_0, \mu_1, \dots, \mu_{n-1}) \in \mathbb{F}_q^{n \times n}$ under the condition that at least two coefficients μ_i and μ_i are different (we suggest to impose all coefficients $\mu_0, \mu_1, \dots, \mu_{n-1}$ to be pair-wise distinct), besides adding proper round constants.

We emphasize that this corresponds to one of the solutions proposed by Vaudenay in [Vau99] (and recently re-considered in [AC21]) for breaking the invariant subspace $\langle [1, 1] \rangle$ of the Lai-Massey scheme over $\mathbb{F}_q^2$. In more details, Vaudenay showed that the invariant subspace can be broken by applying an *orthomorphism* $\mathcal{S}$ on one of the two output of the Lai-Massey scheme, that is, $[x_0, x_1] \mapsto [\mathcal{S}(x_0 + \mathcal{F}(x_0 - x_1)), x_1 + \mathcal{H}(x_0 - x_1)]$. It is easy to check that $x \mapsto \mu \cdot x$ for $\mu \notin \{0, 1\}$ is an orthomorphism (we recall that a function $\mathcal{S}$ is an orthomorphism if and only if both $x \mapsto \mathcal{S}(x)$ and $x \mapsto \mathcal{S}(x) - x$ are permutations).

3.2 Affine Equivalence Relation between Feistel Networks and Lai-Massey Schemes

Next, we prove that an affine equivalence relation holds between the Lai-Massey scheme over $\mathbb{F}_q^n$ proposed in Prop. 1 and the generalized Feistel network.

Proposition 3. *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. The Lai-Massey scheme over $\mathbb{F}_q^n$ defined as in Prop. 1 is affine equivalent to a generalized Feistel network defined in Def. 2.*

Proof. We consider separately the case $n = 2$ and $n \geq 3$. In the case $n \geq 3$, we limit ourselves to prove the results for the two extremes and most commonly used cases, that is,

1. the case $l = 1$ in which the function $\mathcal{F}$ in the Lai-Massey scheme over $\mathbb{F}_q^n$ as proposed in Prop. 1 depends only on a single linear combinations of the inputs
2. the case $l = n - 1$ in which it depends on $n - 1$ independent linear combinations of the inputs.

The other “intermediate” cases can be easily proved by combining the two strategies proposed for these two extreme cases.

In all cases, we prove such result by finding affine invertible transformations A and B over $\mathbb{F}_q^n$ for which the affine equivalence holds (C is always equal to 0 in the following). Since we only deal with linear invertible transformations $A, B : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$, we simply identify them with the corresponding matrices in $\mathbb{F}_q^{n \times n}$.

Case: $n = 2$. Let's start with the case $n = 2$, which concerns the A-equivalence between $[x_0, x_1] \mapsto [x_0 + \mathcal{F}(x_0 - x_1), x_1 + \mathcal{F}(x_0 - x_1)]$ and $[x_0, x_1] \mapsto [x_1 + \mathcal{F}(x_0), x_0]$. In this case, LM over $\mathbb{F}_q^2$ is affine equivalent to the Feistel network F defined as $[x_0, x_1] \mapsto [x_1 + \mathcal{F}(x_0), x_0]$ via the invertible linear transformations

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

and $C = 0$. Indeed:

$$\begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \xrightarrow{A \times \cdot} \begin{bmatrix} x_0 - x_1 \\ x_1 \end{bmatrix} \xrightarrow{F(\cdot)} \begin{bmatrix} x_1 + \mathcal{F}(x_0 - x_1) \\ x_0 - x_1 \end{bmatrix} \xrightarrow{B \times \cdot} \begin{bmatrix} x_0 + \mathcal{F}(x_0 - x_1) \\ x_1 + \mathcal{F}(x_0 - x_1) \end{bmatrix},$$

which is the Lai-Massey scheme. That is, the Lai-Massey scheme is a Feistel network pre- and post-processed with two invertible linear functions.

Case: $n \geq 3$ and $l = 1$ (A-Equivalent to a Type-I Feistel Network). We start by considering a Lai-Massey scheme over $\mathbb{F}_q^n$ as proposed in Prop. 1 for $l = 1$, that is, $x_i \mapsto y_i = x_i + \mathcal{F}\left(\sum_{j=0}^{n-1} \lambda_j \cdot x_j\right)$ for each $i \in \{0, 1, \dots, n-1\}$, where $\mathcal{F} : \mathbb{F}_q \rightarrow \mathbb{F}_q$ and where $\lambda_0, \lambda_1, \dots, \lambda_{n-1} \in \mathbb{F}_q$ satisfy $\sum_{i=0}^{n-1} \lambda_i = 0$.

W.l.o.g., we assume $\lambda_0 \neq 0$.⁴ The analyzed Lai-Massey scheme is affine equivalent to a Type-I Feistel network $F_{\text{Type-I}}$ over $\mathbb{F}_q^n$ defined as

$$[x_0, x_1, x_2, \dots, x_{n-1}] \mapsto [x_1 + \mathcal{F}(x_0), x_2, \dots, x_{n-1}, x_0]$$

via the invertible linear transformations

$$A = \begin{bmatrix} \lambda_0 & \lambda_1 & \lambda_2 & \dots & \lambda_{n-1} \\ 0 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & -1 & 0 & \dots & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \frac{\lambda_2}{\lambda_0} & \dots & -\frac{\lambda_{n-1}}{\lambda_0} & \frac{1}{\lambda_0} \\ 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 1 & 0 & \dots & 1 \end{bmatrix}, \quad (4)$$

and $C = 0$. Indeed, we have that

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{bmatrix} \xrightarrow{A \times \cdot} \begin{bmatrix} \sum_{i=0}^{n-1} \lambda_i \cdot x_i \\ x_1 \\ x_2 - x_1 \\ \vdots \\ x_{n-1} - x_1 \end{bmatrix} \xrightarrow{F_{\text{Type-I}}} \begin{bmatrix} x_1 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i \cdot x_i\right) \\ x_2 - x_1 \\ \vdots \\ x_{n-1} - x_1 \\ \sum_{i=0}^{n-1} \lambda_i \cdot x_i \end{bmatrix} \xrightarrow{B \times \cdot} \begin{bmatrix} x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i \cdot x_i\right) \\ x_1 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i \cdot x_i\right) \\ x_2 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i \cdot x_i\right) \\ \vdots \\ x_{n-1} + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i \cdot x_i\right) \end{bmatrix}.$$

⁴ If $\lambda_0 = 0$, then the following argument works by considering another equivalent Type-I Feistel network (e.g., if $\lambda_i \neq 0$, then it is sufficient to work with $y_i = x_{i+1} + \mathcal{F}(x_{i+2})$ a part from $y_j = x_{j+1}$ for $j = i$).

Case: $n \geq 3$ and $l = n - 1$ (A-Equivalent to a Contracting Feistel Network). Next, we consider the case of a Lai-Massey scheme over $\mathbb{F}_q^n$ as proposed in Prop. 1 for $l = n - 1$ instantiated with $\mathcal{F} : \mathbb{F}_q^{n-1} \rightarrow \mathbb{F}_q$, that is, $x_i \mapsto y_i = x_i + \mathcal{F}(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(n-2)} \cdot x_j)$ for each $i \in \{0, 1, \dots, n-1\}$, where $\lambda_i^{(j)} \in \mathbb{F}_q$ are $n - 1$ “zero-sum linearly independent” sets as in Def. 1.

In order to show that the analyzed Lai-Massey scheme is affine equivalent to a contracting Feistel network F_C defined over $\mathbb{F}_q^n$ as

$$[x_0, x_1, x_2, \dots, x_{n-1}] \mapsto [x_1, x_2, \dots, x_{n-1}, x_0 + \mathcal{F}(x_1, x_2, \dots, x_{n-1})],$$

we introduce the values $\mu_{i,0}, \dots, \mu_{i,n-2} \in \mathbb{F}_q$ for each $i \in \{1, \dots, n-1\}$ defined as the ones that satisfy the following equality:

$$\forall i \in \{1, \dots, n-1\}, \quad \begin{bmatrix} \lambda_0^{(0)} & \lambda_0^{(1)} & \dots & \lambda_0^{(n-2)} \\ \lambda_1^{(0)} & \lambda_1^{(1)} & \dots & \lambda_1^{(n-2)} \\ \vdots & & \ddots & \vdots \\ \lambda_{n-1}^{(0)} & \lambda_{n-1}^{(1)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix} \times \begin{bmatrix} \mu_{i,0} \\ \mu_{i,1} \\ \vdots \\ \mu_{i,n-2} \end{bmatrix} = \begin{bmatrix} -1 \\ \delta_{i,1} \\ \vdots \\ \delta_{i,n-2} \\ \delta_{i,n-1} \end{bmatrix}, \quad (5)$$

where $\delta_{i,j}$ is the Kronecker delta (that is, $\delta_{i,j} = 1$ if $i = j$, and 0 otherwise). The left-hand side (l.h.s.) matrix has $n - 1$ columns and n rows. However, its rows are not linearly independent, since their sum is equal to the zero vector (due to the condition on $\lambda_i^{(j)}$), or equivalently, the sum of each column is equal to zero. Since the right-hand side (r.h.s.) vector satisfies the same zero sum, the previous system of linear equations reduces to

$$\forall i \in \{1, \dots, n-1\}, \quad \begin{bmatrix} \lambda_0^{(0)} & \lambda_0^{(1)} & \dots & \lambda_0^{(n-2)} \\ \lambda_1^{(0)} & \lambda_1^{(1)} & \dots & \lambda_1^{(n-2)} \\ \vdots & & \ddots & \vdots \\ \lambda_{n-2}^{(0)} & \lambda_{n-2}^{(1)} & \dots & \lambda_{n-2}^{(n-2)} \end{bmatrix} \times \begin{bmatrix} \mu_{i,0} \\ \mu_{i,1} \\ \vdots \\ \mu_{i,n-2} \end{bmatrix} = \begin{bmatrix} -1 \\ \delta_{i,1} \\ \vdots \\ \delta_{i,n-2} \end{bmatrix},$$

where the l.h.s. matrix is invertible due to the fact that the vectors $\hat{\lambda}^{(0)}, \hat{\lambda}^{(1)}, \dots, \hat{\lambda}^{(n-2)}$ are linearly independent, as proved in Lemma 6 – see App. A.1.

Given $\mu_{i,j}$ as before, we can now show that the analyzed Lai-Massey scheme is EA-equivalent to a contracting Feistel network F_C defined over $\mathbb{F}_q^n$ via the invertible linear transformations

$$A = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ \lambda_0^{(0)} & \lambda_1^{(0)} & \lambda_2^{(0)} & \dots & \lambda_{n-1}^{(0)} \\ \lambda_0^{(1)} & \lambda_1^{(1)} & \lambda_2^{(1)} & \dots & \lambda_{n-1}^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{(n-2)} & \lambda_1^{(n-2)} & \lambda_2^{(n-2)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ \mu_{1,0} & \mu_{1,1} & \dots & \mu_{1,n-2} & 1 \\ \mu_{2,0} & \mu_{2,1} & \dots & \mu_{2,n-2} & 1 \\ \vdots & & \ddots & \vdots & \vdots \\ \mu_{n-1,0} & \mu_{n-1,1} & \dots & \mu_{n-1,n-2} & 1 \end{bmatrix},$$

and $C = 0$. Indeed, we have that

$$\begin{aligned}
& \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix} \xrightarrow{A \times \cdot} \begin{bmatrix} x_0 \\ \sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i \\ \vdots \\ \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i \end{bmatrix} \xrightarrow{F_C(\cdot)} \begin{bmatrix} \sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i \\ \vdots \\ \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i \\ x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \end{bmatrix} \\
& \xrightarrow{B \times \cdot} \begin{bmatrix} x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \\ \sum_{j=0}^{n-2} \mu_{1,j} \cdot \left(\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i\right) + x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \\ \vdots \\ \sum_{j=0}^{n-2} \mu_{n-1,j} \cdot \left(\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i\right) + x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \end{bmatrix} \\
& = \begin{bmatrix} x_0 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \\ x_1 + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \\ \vdots \\ x_{n-1} + \mathcal{F}\left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(n-2)} \cdot x_i\right) \end{bmatrix},
\end{aligned}$$

where the last equality holds due to the definition of $\mu_{i,j}$ – see App. A.2 for more details. $\square$

Interestingly, the affine equivalences just given also hold when considering $r \geq 2$ rounds of the Lai-Massey construction and of the Feistel scheme (*without the rotation* at the end of each round).

Indeed, let us focus on the $\mathbb{F}_q^2$ -case. By considering $r \geq 2$ consecutive rounds of the Lai-Massey construction, we get the following

$$\mathcal{LM}^r(x) = (B \times \mathcal{F} \circ A)^r(x) = A^{-1} \times (A \times B \times \mathcal{F})^r(A \times x),$$

where

$$A \times B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \text{and}$$

$$A \times B \times \mathcal{F}(x_0, x_1) = [x_0, x_1 + F(x_0)]$$

correspond to the Feistel scheme over $\mathbb{F}_q^2$ without the swapping. The conclusion holds over $\mathbb{F}_q^n$ for $n \geq 3$ as well – in order to prove it, it is sufficient to check that $A \times (B \times \text{circ}(0, 1, 0, \dots, 0))$ is equal to the identity matrix.

For this reason, we can conclude the following.

Proposition 4. *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. The r -round Lai-Massey scheme over $\mathbb{F}_q^n$ defined as in Prop. 1 is affine equivalent to a r -round generalized Feistel network defined in Def. 2 without the rotation.*

As a side result, this also implies the existence of invariant subspaces for the Lai-Massey scheme over $\mathbb{F}_q^n$ defined as in Prop. 1, as already pointed out before.

3.3 The Unified Structure

Recently, Liu et al. [LSL⁺23] and Sun et al. [SXD⁺24] proposed the *unified structure*, which allows to describe both Feistel-like networks and Lai-Massey schemes that are instantiated via a single function $\mathcal{F}$ within a single framework.

Proposition 5 (Unified Structure). *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Let $\lambda_0, \lambda_1, \dots, \lambda_{n-1} \in \mathbb{F}_q$ and $\beta_0, \beta_1, \dots, \beta_{n-1} \in \mathbb{F}_q$ be such that $\sum_{i=0}^{n-1} \lambda_i \cdot \beta_i = 0$. Let $\mathcal{F} : \mathbb{F}_q \rightarrow \mathbb{F}_q$ be a function.*

The unified structure US defined as $\text{US}(x_0, x_1, \dots, x_{n-1}) := (y_0, y_1, \dots, y_{n-1})$ where

$$y_i := x_i + \beta_i \cdot \mathcal{F}\left(\sum_{j=0}^{n-1} \lambda_j \cdot x_j\right)$$

for each $i \in \{0, 1, \dots, n-1\}$ is invertible.

As in any Lai-Massey scheme, the invertibility follows from the fact that it is always possible to recover the linear combination $\sum_{j=0}^{n-1} \lambda_j \cdot x_j$ as following:

$$\sum_{i=0}^{n-1} \lambda_i \cdot y_i = \sum_{i=0}^{n-1} \lambda_i \cdot x_i + \mathcal{F} \left(\sum_{j=0}^{n-1} \lambda_j \cdot x_j \right) \cdot \underbrace{\sum_{i=0}^{n-1} \lambda_i \cdot \beta_i}_{=0} = \sum_{i=0}^{n-1} \lambda_i \cdot x_i.$$

Inspired by [GØSW23], here we propose a generalized version of the unified structure just recalled.

Proposition 6. *Let $n \geq 2$ be an integer, and let $q = p^s$ be as before. Let $\alpha_0, \alpha_1, \dots, \alpha_{n-1} \in \mathbb{F}_p \setminus \{0\}$. Let $l \in \{1, 2, \dots, n-1\}$. Let $\mathcal{F} : \mathbb{F}_q^l \rightarrow \mathbb{F}_q$ be a function. Let $\beta_0, \beta_1, \dots, \beta_{n-1} \in \mathbb{F}_q$, and let $\{\lambda_j^{(i)}\}_{j \in \{0, 1, \dots, n-1\}, i \in \{0, 1, \dots, l-1\}}$ with $\lambda_j^{(i)} \in \mathbb{F}_q$ be l sets such that*

- *for each $i \in \{0, 1, \dots, l-1\}$: $\sum_{j=0}^{n-1} \lambda_j^{(i)} \cdot \beta_j = 0$;*
- *for each distinct $i, i' \in \{0, 1, \dots, l-1\}$: $\{\lambda_j^{(i)}\}_j$ and $\{\lambda_j^{(i')}\}_j$ are linearly independent.*

The generalized unified structure GUS over $\mathbb{F}_q^n$ defined as $\text{GUS}(x_0, x_1, \dots, x_{n-1}) := y_0 \| y_1 \| \dots \| y_{n-1}$ where

$$y_i := \alpha_i \cdot \left(x_i + \beta_i \cdot \mathcal{F} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right) \right)$$

for each $i \in \{0, 1, \dots, n-1\}$ is invertible.

The invertibility of this structure is completely analogous of the one just given for the unified structure US. We emphasize that the generalized unified structure reduces to the Lai-Massey scheme proposed in Prop. 1 for $\beta_0 = \beta_1 = \dots = \beta_{n-1} = 1$. In particular, the set $\mathfrak{X}$ defined in (3) is invariant for the generalized unified structure.

Working as in Sect. 3.2, it is easy to check that any Feistel scheme and any Lai-Massey scheme are affine equivalent to the generalized unified structure just proposed. In the next section, we will show how to construct a scheme that is *not* affine equivalent to the Feistel/Lai-Massey scheme, by considering the Lai-Massey scheme proposed in Prop. 1 as the starting point for the rest of the paper.

4 Generalized Lai-Massey Schemes

As next step, we discuss possible generalizations of the Lai-Massey scheme. Our goal is to introduce schemes that (i) capture the main idea of the Lai-Massey scheme, and that (ii) are not EA-equivalent to any Feistel network. (In the following, we denote the “EA-equivalence class” – or “EA-class” for brevity – of generalized Feistel networks as “Feistel EA-class”.) In this section, we focus on the “generalized Lai-Massey” schemes, while the “redundant Lai-Massey” schemes is discussed in the next one.

4.1 Definition of Generalized Lai-Massey Schemes

One main feature of the Lai-Massey scheme proposed in Prop. 1 regards the fact that the inputs of the function $\mathcal{F}$ are linear combinations of the inputs x_i defined via coefficients $\lambda_j^{(i)}$ that sum to zero (that is, $\sum_{j=0}^{n-1} \lambda_j^{(i)} = 0$ for each $i \in \{0, 1, \dots, l-1\}$). A possible generalization of such design consists in allowing for n different functions $\mathcal{F}_0, \mathcal{F}_1, \dots, \mathcal{F}_{n-1}$, under the restriction that their inputs are zero-sum linear combinations of x_i as before. More formally:

Definition 5 (Generalized Lai-Massey). *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Let $l \in \{1, 2, \dots, n-1\}$. Let $\{\lambda_j^{(i)}\}_{j \in \{0, 1, \dots, n-1\}, i \in \{0, 1, \dots, l-1\}}$ be l “zero-sum linearly independent” sets as in Def. 1. Given n functions $\mathcal{F}^{(0)}, \mathcal{F}^{(1)}, \dots, \mathcal{F}^{(n-1)} : \mathbb{F}_q^l \rightarrow \mathbb{F}_q$, let $\text{LM}_G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ be defined as*

$$\text{LM}_G(x_0, x_1, \dots, x_{n-1}) = y_0 \| y_1 \| y_2 \| \dots \| y_{n-1}$$

where

$$y_i := x_i + \mathcal{F}^{(i)} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right)$$

for each $i \in \{0, 1, \dots, n-1\}$. We say that LM_G is a generalized Lai-Massey scheme if it is invertible.

Obviously, the Lai-Massey scheme defined in Prop. 1 satisfies this definition. Other examples of invertible generalized Lai-Massey constructions over $\mathbb{F}_q^4$ (analogous over $\mathbb{F}_q^n$ for $n \geq 2$) include

$$\begin{aligned} [y_0, y_1, y_2, y_3] &= [x_0 + \mathcal{F}(x_0 - x_1), x_1 + \mathcal{F}(x_0 - x_1), x_2, x_3], \\ [y_0, y_1, y_2, y_3] &= [x_0 + \mathcal{F}(x_0 - x_1), x_1 + \mathcal{F}(x_0 - x_1), x_2 + \mathcal{F}'(x_2 - x_3), x_3 + \mathcal{F}'(x_2 - x_3)], \end{aligned}$$

where $\mathcal{F}, \mathcal{F}' : \mathbb{F}_q \rightarrow \mathbb{F}_q$.

4.2 A Generalized Lai-Massey Scheme *Not* Belonging into the “Feistel EA-Class”

Next, we propose a concrete example of a generalized Lai-Massey scheme as in Def. 5 over $\mathbb{F}_q^n$ that is *not* EA-equivalent to any generalized Feistel network. We first propose it over $\mathbb{F}_q^4$ in Prop. 7, and then we iteratively generalize it over $\mathbb{F}_q^n$ for each $n = 2 \cdot n' \geq 6$ even.⁵

Proposition 7 (GLM₄). *Given $q = p^s$ as before. For each $i \in \{1, 2, 3\}$, let $\mathcal{F}_i : \mathbb{F}_q^i \rightarrow \mathbb{F}_q$ be a function. The generalized Lai-Massey scheme $\text{GLM}_4(x_0, x_1, x_2, x_3) = y_0 \| y_1 \| y_2 \| y_3$ over $\mathbb{F}_q^4$ defined as*

$$\begin{aligned} y_0 &:= x_0 + \mathcal{F}_1(x_0 - x_1) + \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3), \\ y_1 &:= x_1 + \mathcal{F}_1(x_0 - x_1) + \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3), \\ y_2 &:= x_2 + \mathcal{F}_2(x_0 - x_1, x_2 - x_3) + \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3), \\ y_3 &:= x_3 + \mathcal{F}_2(x_0 - x_1, x_2 - x_3) + \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3); \end{aligned}$$

is invertible.

Proof. Working as for any Lai-Massey scheme, the invertibility follows from the facts:

$$\begin{aligned} x_0 - x_1 &= y_0 - y_1, & x_2 - x_3 &= y_2 - y_3, \\ x_1 - x_2 &= y_1 - y_2 - \mathcal{F}_1(y_0 - y_1) - \mathcal{F}_2(y_0 - y_1, y_2 - y_3). \end{aligned}$$

By making use of the same strategy exploited for the Lai-Massey scheme e.g. in Prop. 1, these information are sufficient for recovering x_0, x_1, x_2, x_3 . $\square$

Proposition 8 (GLM_n). *Let $q = p^s$ be as before, and let $n = 2 \cdot n' \geq 6$. For each $i \in \{1, 2, \dots, n-1\}$, let $\mathcal{F}_i : \mathbb{F}_q^i \rightarrow \mathbb{F}_q$ be a function.*

For each even integer $n = 2 \cdot n' \geq 6$, the generalized Lai-Massey scheme $\text{GLM}_n(x_0, x_1, \dots, x_{n-1}) = y_0 \| y_1 \| \dots \| y_{n-1}$ over $\mathbb{F}_q^n$ defined as

$$y_i := \begin{cases} z_i + \mathcal{F}_{n-1}(w_0, w_1, \dots, w_{n-4}, w_{n-3}, w_{n-2}) & \text{if } i \in \{0, 1, \dots, n-3\}, \\ x_i + \mathcal{F}_{n-2}(w_0, w_1, \dots, w_{n-4}, w_{n-2}) & \text{otherwise} \\ \quad + \mathcal{F}_{n-1}(w_0, w_1, \dots, w_{n-4}, w_{n-3}, w_{n-2}) \end{cases}$$

where (i) $w_i := x_i - x_{i+1}$, and (ii) $[z_0, z_1, \dots, z_{n-3}] := \text{GLM}_{n-2}(x_0, x_1, \dots, x_{n-3})$ is the output of the generalized Lai-Massey scheme GLM_{n-2} over $\mathbb{F}_q^{n-2}$, is invertible.

The proof – analogous to the one for the case $n = 4$ – is given in App. B.

As before, we point out that any difference in the subspace $\langle [1, 1, \dots, 1] \rangle \equiv \{[x, x, \dots, x] \mid x \in \mathbb{F}_q\} \subseteq \mathbb{F}_q^n$ does not activate any function $\mathcal{F}_i$ of the generalized Lai-Massey schemes GLM_n . It is not hard to check that such subspace can be broken by exploiting the results proposed in Sect. 3.1. In particular, since we are in the case $l = n - 1$, such result can be achieved by multiplying each y_i by a non-null coefficient μ_i under the conditions that the coefficients $\mu_0, \mu_1, \dots, \mu_{n-1}$ are pair-wise different.

⁵ We limit ourselves to mention that it is possible to set up an analogous invertible scheme that is *not* EA-equivalent to any generalized Feistel network over $\mathbb{F}_q^n$ for each $n = 2 \cdot n' + 1 \geq 5$ odd as well.

About the EA-Equivalence. The generalized Lai-Massey schemes just proposed in Prop. 7 – 8 are not EA-equivalent to any generalized Feistel network.

Proposition 9. *Let $q = p^s$ be as before, and let $n \geq 4$. The generalized Lai-Massey constructions GLM_n proposed in Prop. 7 – 8 are **not** extended affine equivalent to any generalized Feistel network.*

Proof. Let's start the proof by considering the case $n = 4$. If the EA-equivalence holds, then there must exist invertible affine layers A, B and an affine layer C over $\mathbb{F}_q^4$ such that $\text{GLM}_4(x) = B \circ \text{GF} \circ A(x) + C(x)$, where GF is defined in Def. 2. Let's first consider the case in which A, B, C are linear. Since GLM_4 depends on $x_0 - x_1, x_1 - x_2, x_2 - x_3$, then the invertible matrix A must be of the form

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ \psi_0 & \psi_1 & \psi_2 & \psi_3 \end{bmatrix},$$

where $\psi_0, \psi_1, \psi_2, \psi_3 \in \mathbb{F}_q$ must satisfy $\det(A) = -(\psi_0 + \psi_1 + \psi_2 + \psi_3) \neq 0$. Given A , we have that

$$\text{GF} \circ A(x) = \begin{bmatrix} x_2 - x_3 + \mathcal{F}_1(x_0 - x_1) \\ x_1 - x_2 + \mathcal{F}_2(x_0 - x_1, x_2 - x_3) \\ \psi_0 \cdot x_0 + \psi_1 \cdot x_1 + \psi_2 \cdot x_2 + \psi_3 \cdot x_3 + \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3) \\ x_0 - x_1 \end{bmatrix}.$$

In order to realize the EA-equivalence, the matrix B must be of the form

$$B = \begin{bmatrix} \varphi_0 & 0 & \varphi_2 & \varphi_3 \\ \varphi_0 & 0 & \varphi_2 & \varphi_4 \\ 0 & \varphi_1 & \varphi_2 & \varphi_5 \\ 0 & \varphi_1 & \varphi_2 & \varphi_6 \end{bmatrix} \quad (6)$$

for $\varphi_0, \varphi_1, \dots, \varphi_6 \in \mathbb{F}_q$. Indeed, due to the distribution of the functions $\mathcal{F}_i$ in GLM_4 , we have that $B \circ \text{GF} \circ A(x)$ is equal to

$$\begin{bmatrix} L^{(0)}(x_0, x_1, x_2, x_3) + \varphi_0 \cdot \mathcal{F}_1(x_0 - x_1) + \varphi_2 \cdot \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3) \\ L^{(1)}(x_0, x_1, x_2, x_3) + \varphi_0 \cdot \mathcal{F}_1(x_0 - x_1) + \varphi_2 \cdot \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3) \\ L^{(2)}(x_0, x_1, x_2, x_3) + \varphi_1 \cdot \mathcal{F}_2(x_0 - x_1, x_2 - x_3) + \varphi_2 \cdot \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3) \\ L^{(3)}(x_0, x_1, x_2, x_3) + \varphi_1 \cdot \mathcal{F}_2(x_0 - x_1, x_2 - x_3) + \varphi_2 \cdot \mathcal{F}_3(x_0 - x_1, x_1 - x_2, x_2 - x_3) \end{bmatrix},$$

where $L^{(i)} : \mathbb{F}_q^4 \rightarrow \mathbb{F}_q$ is a linear function for each $i \in \{0, 1, 2, 3\}$. Independently of the value of φ_i , the matrix B is never invertible, since the first three columns are always linearly dependent:

$$\forall i \in \{0, 1, 2, 3\} : \quad \varphi_1 \cdot \varphi_2 \cdot B_{i,0} + \varphi_0 \cdot \varphi_2 \cdot B_{i,1} - \varphi_0 \cdot \varphi_1 \cdot B_{i,2} = 0.$$

The same conclusion holds when considering affine layers A, B over $\mathbb{F}_q^4$. We point out that the affine layer C only affects the linear combination of the inputs x_0, x_1, x_2, x_3 , hence, it does not affect the previous result. It follows that no EA-equivalence holds over $\mathbb{F}_q^4$.

The proof for the case $n = 2 \cdot n' \geq 6$ even is analogous to the one given for the case $n = 4$. Again, the EA-equivalence does not hold due to the fact that the matrix B is not invertible. By working as before, it is possible to construct an invertible matrix $A \in \mathbb{F}_q^{n \times n}$ that returns all the combinations $x_i - x_{i+1}$ for each $i \in \{0, 1, \dots, n-2\}$. Let $B^{(4)} \in \mathbb{F}_q^{4 \times 3}$ be the matrix corresponding to the first three columns of $B \in \mathbb{F}_q^{4 \times 4}$ as defined in (6). For each $n = 2 \cdot n' \geq 6$, we define $B \in \mathbb{F}_q^{n \times n}$ via $B^{(n)} \in \mathbb{F}_q^{n \times (n-1)}$ as

$$B = \left[\begin{array}{c|c} B^{(n)} & \begin{bmatrix} \varphi_{n-1} \\ \varphi_n \\ \vdots \\ \varphi_{2n-2} \\ \varphi_{2n-1} \end{bmatrix} \end{array} \right], \quad B^{(n)} := \left[\begin{array}{c|cc} B^{(n-2)} & 0 & \varphi_{n-2} \\ & \vdots & \vdots \\ & 0 & \varphi_{n-2} \\ \hline 0 & \dots & 0 & \varphi_{n-3} & \varphi_{n-2} \\ 0 & \dots & 0 & \varphi_{n-3} & \varphi_{n-2} \end{array} \right].$$

It is easy to check that the columns of $B^{(n)}$ are linearly dependent due to the fact that the columns of $B^{(n-2)}$ are linearly dependent. As before, B is never invertible since the columns of $B^{(n)}$ are linearly dependent. $\square$

5 Redundant Lai-Massey Schemes

5.1 Definition of Redundant Lai-Massey Schemes

Focusing again on Prop. 1, another main feature of such Lai-Massey schemes $[x_0, x_1, \dots, x_{n-1}] \mapsto [y_0, y_1, \dots, y_{n-1}]$ regards the fact that each output y_i is defined as the sum of the corresponding input x_i and of a certain element $z = \mathcal{F}(x_0, x_1, \dots, x_{n-1})$, that is, $y_i = x_i + z$ where z is fixed for each $i \in \{0, 1, \dots, n-1\}$. In the original Lai-Massey scheme, the inputs of the function $\mathcal{F}$ must be of a particular form (that is, zero-sum linear combinations of $x_0, x_1, \dots, x_{n-1}$) in order to guarantee the invertibility. In the following definition, we remove such a restriction, allowing for a more general scheme.

Definition 6 (Redundant Lai-Massey). *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Given a function $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$, let $\text{LM}_R : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ be defined as*

$$\text{LM}_R(x_0, x_1, \dots, x_{n-1}) = y_0 \| y_1 \| y_2 \| \dots \| y_{n-1}$$

where

$$y_i := x_i + \mathcal{F}(x_0, x_1, \dots, x_{n-1})$$

for each $i \in \{0, 1, \dots, n-1\}$. We say that LM_R is a redundant⁶ Lai-Massey scheme if it is invertible.

Obviously, the Lai-Massey scheme defined in Prop. 1 satisfies this definition.

5.2 A Redundant Lai-Massey Scheme Belonging into the “Feistel EA-Class”

As a first example, we show an invertible redundant Lai-Massey scheme, in which the inputs of the function $\mathcal{F}$ are not necessarily zero-sum linear combinations of $x_0, x_1, \dots, x_{n-1}$.

Lemma 1. *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Let $\mu_0, \mu_1, \dots, \mu_{n-1} \in \mathbb{F}_q$ be such that $\sum_{i=0}^{n-1} \mu_i \neq 0$. Let $\mathcal{H} : \mathbb{F}_q \rightarrow \mathbb{F}_q$ be a permutation. The redundant Lai-Massey scheme $[x_0, x_1, \dots, x_{n-1}] \mapsto [y_0, y_1, \dots, y_{n-1}]$ over $\mathbb{F}_q^n$ defined as*

$$y_i = x_i + \frac{1}{\sum_{j=0}^{n-1} \mu_j} \cdot \left(\mathcal{H} \left(\sum_{j=0}^{n-1} \mu_j \cdot x_j \right) - \sum_{j=0}^{n-1} \mu_j \cdot x_j \right)$$

for each $i \in \{0, 1, \dots, n-1\}$ is invertible.

Proof. The invertibility is a consequence of $\sum_{j=0}^{n-1} \mu_j \cdot x_j = \mathcal{H}^{-1} \left(\sum_{j=0}^{n-1} \mu_j \cdot y_j \right)$, which follows from

$$\begin{aligned} \sum_{i=0}^{n-1} \mu_i \cdot y_i &= \sum_{i=0}^{n-1} \mu_i \cdot x_i + \sum_{i=0}^{n-1} \left(\mu_i \cdot \left(\frac{1}{\sum_{j=0}^{n-1} \mu_j} \cdot \mathcal{H} \left(\sum_{j=0}^{n-1} \mu_j \cdot x_j \right) - \frac{\sum_{j=0}^{n-1} \mu_j \cdot x_j}{\sum_{j=0}^{n-1} \mu_j} \right) \right) \\ &= \sum_{i=0}^{n-1} \mu_i \cdot x_i + \left(\frac{1}{\sum_{j=0}^{n-1} \mu_j} \cdot \mathcal{H} \left(\sum_{j=0}^{n-1} \mu_j \cdot x_j \right) - \frac{\sum_{j=0}^{n-1} \mu_j \cdot x_j}{\sum_{j=0}^{n-1} \mu_j} \right) \cdot \left(\sum_{i=0}^{n-1} \mu_i \right) \\ &= \sum_{i=0}^{n-1} \mu_i \cdot x_i + \mathcal{H} \left(\sum_{j=0}^{n-1} \mu_j \cdot x_j \right) - \sum_{j=0}^{n-1} \mu_j \cdot x_j \\ &= \mathcal{H} \left(\sum_{j=0}^{n-1} \mu_j \cdot x_j \right) \implies \sum_{j=0}^{n-1} \mu_j \cdot x_j = \mathcal{H}^{-1} \left(\sum_{j=0}^{n-1} \mu_j \cdot y_j \right), \end{aligned}$$

⁶ For differentiating it from the previous generalized Lai-Massey scheme, we decided to call this one as “redundant” Lai-Massey scheme to capture the fact that the same function $\mathcal{F}(x_0, x_1, \dots, x_{n-1})$ is repeatedly used for building/defining the output.

since $\mathcal{H}$ is invertible. As a result, for each $i \in \{0, 1, \dots, n-1\}$:

$$x_i = y_i + \frac{1}{\sum_{j=0}^{n-1} \mu_j} \cdot \left(\mathcal{H}^{-1} \left(\sum_{j=0}^{n-1} \mu_j \cdot y_j \right) - \sum_{j=0}^{n-1} \mu_j \cdot y_j \right).$$

□

Even if the inputs of the function $\mathcal{F}$ are not zero-sum linear combinations of $x_0, x_1, \dots, x_{n-1}$, the obtained scheme is EA-equivalent to a contracting Feistel network, due to the same argument proposed in Sect. 3.2. In particular, assuming $\mu_0 \neq 0$, the affine equivalence holds via the invertible matrices $A, B \in \mathbb{F}_q^{n \times n}$ equal to the ones given in (4), while the linear transformation C is defined via the matrix $C \in \mathbb{F}_q^{n \times n}$ identically equal to zero except for $C_{0,1} = -(\sum_{j=0}^{n-1} \mu_j)/\mu_0$.

5.3 A Redundant Lai-Massey Scheme *Not* Belonging into the “Feistel EA-Class”

Next, we propose an example of a redundant Lai-Massey scheme as in Def. 6 over $\mathbb{F}_q^n$ that is *not* EA-equivalent to any generalized Feistel network.

Proposition 10 (RLM). *Let $p \geq 3$ be a prime integer, and let $n \geq 2$. Let $l \in \{1, 2, \dots, n-1\}$. Assume the following conditions:*

- let $\{\lambda_j^{(i)}\}_{j \in \{0,1,\dots,n-1\}, i \in \{0,1,\dots,l-1\}}$ be l “zero-sum linearly independent” sets as in Def. 1;
- let $\mathcal{G} : \mathbb{F}_p^l \rightarrow \mathbb{F}_p$ be a function;
- let $\psi_0, \psi_1, \dots, \psi_{n-1} \in \mathbb{F}_p$ (**no** condition on $\sum_{j=0}^{n-1} \psi_j$), and let $\beta \in \mathbb{F}_p \setminus \{0\}$.
If $\sum_{j=0}^{n-1} \psi_j \neq 0$, we assume that $-\beta \cdot \left(\sum_{j=0}^{n-1} \psi_j \right)$ is a quadratic **non-residue**, that is, $-\beta \cdot \left(\sum_{j=0}^{n-1} \psi_j \right) \neq w^2 \pmod p$ for each $w \in \mathbb{F}_p$.

The redundant Lai-Massey scheme RLM over $\mathbb{F}_p^n$ defined as $\text{RLM}(x_0, x_1, \dots, x_{n-1}) = y_0 \| y_1 \| \dots \| y_{n-1}$ where

$$y_i := x_i + \beta \cdot (z_{x_0, x_1, \dots, x_{n-1}})^2 \cdot \left(\sum_{j=0}^{n-1} \psi_j \cdot x_j \right)$$

for each $i \in \{0, 1, \dots, n-1\}$ with

$$z_{x_0, x_1, \dots, x_{n-1}} \equiv z := \mathcal{G} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right)$$

is invertible.

Proof. If $\sum_{j=0}^{n-1} \psi_j = 0 \pmod p$, then the invertibility follows from Prop. 1. Hence, we focus on the case $\sum_{j=0}^{n-1} \psi_j \neq 0 \pmod p$. Similarly to before, in order to invert it, it is sufficient to find the linear combinations of x_i both with respect to $\lambda_i^{(j)}$ and with respect to ψ_i . Given $y_0, y_1, \dots, y_{n-1}$ as before, we have

$$\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot y_i = \sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i + \beta \cdot \underbrace{\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot z^2}_{=0} \cdot \left(\sum_{j=0}^{n-1} \psi_j \cdot x_j \right) = \sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i$$

for each $j \in \{0, 1, \dots, l-1\}$, where $\sum_{i=0}^{n-1} \lambda_i^{(j)} = 0$ by assumption. It follows that $z = \mathcal{G}(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot y_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot y_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot y_j)$.

If $z = 0$, then $x_i = y_i$ for each $i \in \{0, 1, \dots, n-1\}$. Otherwise, if $z \neq 0$, note that

$$\sum_{j=0}^{n-1} \psi_j \cdot y_j = \left(\sum_{j=0}^{n-1} \psi_j \cdot x_j \right) \cdot \left(1 + \beta \cdot z^2 \cdot \left(\sum_{j=0}^{n-1} \psi_j \right) \right).$$

Such equality is invertible with respect to $\sum_{j=0}^{n-1} \psi_j \cdot x_j$ if $1 + \beta \cdot z^2 \cdot \left(\sum_{j=0}^{n-1} \psi_j\right) \neq 0$, that is, $-\beta \cdot \left(\sum_{j=0}^{n-1} \psi_j\right) \neq 1/z^2$ for each $z \in \mathbb{F}_p$. This condition is always satisfied under the assumption that $-\beta \cdot \left(\sum_{j=0}^{n-1} \psi_j\right)$ is a quadratic **non**-residue mod p . Given both z and $\sum_{j=0}^{n-1} \psi_j \cdot x_j$, it is trivial to invert the system. $\square$

As before, we point out that the scheme RLM admits invariant subspaces. In particular, the scheme is linear over the subspace $\{[x_0, x_1, \dots, x_{n-1}] \in \mathbb{F}_p^n \mid \sum_{j=0}^{n-1} \psi_j \cdot x_j = 0\} \subseteq \mathbb{F}_p^n$. Moreover, any difference in the subspace $\langle [1, 1, \dots, 1] \rangle \subseteq \mathbb{F}_p^n$ does not active the function $\mathcal{G}$. Similar to before, any invariant subspace can be broken by exploiting the results proposed in Sect. 3.1.

About the EA-Equivalence. Here we show that the redundant Lai-Massey scheme just defined is not EA-equivalent to any generalized Feistel network.

Proposition 11. *Let $p \geq 3$ be a prime integer, and let $n \geq 2$. The redundant Lai-Massey scheme RLM defined in Prop. 10 for which (i) $\sum_{j=0}^{n-1} \psi_j \neq 0 \pmod{p}$, and (ii) $l = n - 1$ (where $\mathcal{G}$ is a non-trivial function depending on $l = n - 1$ inputs), is not EA-equivalent to any generalized Feistel network.*

Proof. The proof follows from the facts that

- the functions $\mathcal{F}_i$ in any generalized Feistel network as in Def. 2 depend on at most $i \leq n - 1$ linearly-independent inputs;
- the function $\mathcal{F}(x_0, \dots, x_{n-1}) := \beta \cdot z^2 \cdot \left(\sum_{j=0}^{n-1} \psi_j \cdot x_j\right)$ in Prop. 10 (where $z := \mathcal{G}(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(n-2)} \cdot x_j)$) depends on n linearly-independent inputs.

As a result, the equivalence $\text{RLM}(x) = B \circ \text{GF} \circ A(x) + C(x)$ is never realized for any invertible affine layer A, B and for any affine layer C . $\square$

6 The Blooming of the Amaryllises Construction

In this section, we present a new variant of the Lai-Massey scheme called **Amaryllises** that takes inspiration from the **Horst** construction previously recalled.

6.1 The Amaryllises Construction

The main feature of a **Horst** construction regards the fact that the linear combination that takes place in a Feistel network can be replaced by a non-linear combination. As concretely showed in [GHR⁺23] (and recalled in the following), such a construction can have some concrete advantages with respect to the Feistel networks when the goal is to guarantee security against algebraic attacks in an efficient way.

Here, we apply a similar design strategy on a Lai-Massey scheme in order to set up a construction – called **Amaryllises** – with similar advantages. Our result is presented in the following Theorem.

Theorem 2 (Amaryllises). *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Assume the following conditions:*

- let $\{\lambda_j^{(i)}\}_{j \in \{0,1,\dots,n-1\}, i \in \{0,1,\dots,l-1\}}$ be l “zero-sum linearly independent” sets as in Def. 1;
- let $\mathcal{H} : \mathbb{F}_q^l \rightarrow \mathbb{F}_q$ be a function;
- let $\mathcal{F} : \mathbb{F}_q \rightarrow \mathbb{F}_q$ be a function such that (1st) $\mathcal{F}(0) \neq 0$ and (2nd) $\mathcal{G}(x) := x \cdot \mathcal{F}(x)$ is invertible over $\mathbb{F}_q$;
- let $\beta_0, \beta_1, \dots, \beta_{n-1} \in \mathbb{F}_q \setminus \{0\}$ such that $\sum_{i=0}^{n-1} \beta_i = 0$ **if** $\mathcal{H}$ is not identically equal to zero (equivalently, no condition on $\sum_{i=0}^{n-1} \beta_i$ is imposed if $\mathcal{H}(z) = 0$ for each $z \in \mathbb{F}_q$).

The **Amaryllises** construction **A** over $\mathbb{F}_q^n$ defined as $\mathbf{A}(x_0, \dots, x_{n-1}) = y_0 \parallel \dots \parallel y_{n-1}$ where

$$y_i := x_i \cdot \mathcal{F} \left(\sum_{j=0}^{n-1} \beta_j \cdot x_j \right) + \mathcal{H} \left(\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot x_j, \sum_{j=0}^{n-1} \lambda_j^{(1)} \cdot x_j, \dots, \sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot x_j \right) \quad (7)$$

for each $i \in \{0, 1, \dots, n - 1\}$ is invertible.

Proof. Firstly, we prove that $\mathcal{F}(z) \neq 0$ for each $z \in \mathbb{F}_q$. Since $\mathcal{G}$ is a permutation and since $\mathcal{G}(0) = \mathcal{F}(0) \cdot 0 = 0$ by definition, then $\mathcal{G}(x) \neq 0$ for each $x \neq 0$. It follows that $\mathcal{F}(x) = \mathcal{G}(x)/x \neq 0$ for any $x \in \mathbb{F} \setminus \{0\}$, while $\mathcal{F}(0) \neq 0$ by assumption.

As before, the inverse of $\mathbf{A}$ can be constructed once the linear combinations of y_i with respect to β_i and to $\lambda_i^{(j)}$ are known. Given $y_0, y_1, \dots, y_{n-1}$, it is possible to recover $\sum_{i=0}^{n-1} \beta_i \cdot x_i$ by noting the following:

$$\begin{aligned} \sum_{i=0}^{n-1} \beta_i \cdot y_i &= \left(\sum_{i=0}^{n-1} \beta_i \cdot x_i \right) \cdot \mathcal{F} \left(\sum_{i=0}^{n-1} \beta_i \cdot x_i \right) \\ &\quad + \underbrace{\mathcal{H} \left(\sum_{i=0}^{n-1} \lambda_i^{(0)} \cdot x_i, \sum_{i=0}^{n-1} \lambda_i^{(1)} \cdot x_i, \dots, \sum_{i=0}^{n-1} \lambda_i^{(l-1)} \cdot x_i \right)}_{=0} \cdot \sum_{i=0}^{n-1} \beta_i \\ &= \mathcal{G} \left(\sum_{i=0}^{n-1} \beta_i \cdot x_i \right) \end{aligned}$$

which implies

$$\sum_{i=0}^{n-1} \beta_i \cdot x_i = \mathcal{G}^{-1} \left(\sum_{i=0}^{n-1} \beta_i \cdot y_i \right),$$

where remember that $\mathcal{G}$ is invertible by definition. Note that either $\mathcal{H}$ always returns zero (that is, $\mathcal{H}(x) = 0$ for each $x \in \mathbb{F}_q$) or $\sum_{i=0}^{n-1} \beta_i = 0$ by assumption.

Similarly, since $\sum_{i=0}^{n-1} \lambda_i^{(j)} = 0$, it is possible to recover $\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i$ for each $j \in \{0, 1, \dots, l-1\}$:

$$\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot x_i = \frac{1}{\mathcal{F} \left(\mathcal{G}^{-1} \left(\sum_{i=0}^{n-1} \beta_i \cdot y_i \right) \right)} \cdot \left(\sum_{i=0}^{n-1} \lambda_i^{(j)} \cdot y_i \right),$$

where $\mathcal{F} \left(\mathcal{G}^{-1} \left(\sum_{i=0}^{n-1} \beta_i \cdot y_i \right) \right) \neq 0$ (remember that $\mathcal{F}$ never returns zero).

Given $z := \mathcal{F} \left(\mathcal{G}^{-1} \left(\sum_{i=0}^{n-1} \beta_i \cdot y_i \right) \right)$, it follows that

$$x_i = \frac{1}{z} \cdot \left(y_i - \mathcal{H} \left(\frac{\sum_{j=0}^{n-1} \lambda_j^{(0)} \cdot y_j}{z}, \dots, \frac{\sum_{j=0}^{n-1} \lambda_j^{(l-1)} \cdot y_j}{z} \right) \right) \quad (8)$$

for each $i \in \{0, \dots, n-1\}$. $\square$

Multiplicative Complexity of Amaryllises. Before going on, we point out that the cost of computing **Amaryllises** corresponds to (i) the cost of computing $\mathcal{F}$ and $\mathcal{H}$, besides (ii) n (non-linear) $\mathbb{F}_q$ -multiplications⁷ and n $\mathbb{F}_q$ -sums.

6.2 Suitable Functions for the Amaryllises Construction

As next step, we show how to construct functions $\mathcal{F}$ that satisfy the required assumptions of the previous Theorem 2.

Lemma 2. *Let $q = p^s$ be as before. Let $\mathcal{P}$ be a permutation over $\mathbb{F}_q$. Let $\psi \in \mathbb{F}_q \setminus \{0\}$. The function $\mathcal{F}$ over $\mathbb{F}_q$ defined as*

$$\mathcal{F}(x) := \begin{cases} \frac{\mathcal{P}(x) - \mathcal{P}(0)}{x} & \text{if } x \neq 0 \\ \psi & \text{otherwise } (x = 0) \end{cases}$$

satisfies the requirements of Theorem 2.

The proof is trivial (note that $\mathcal{F}(0) = \psi \neq 0$, and $x \mapsto x \cdot \mathcal{F}(x)$ is a permutation). By exploiting this result, a concrete example of functions $\mathcal{F}$ that satisfy the assumptions of Theorem 2 and that are cheap from the point of view of the *multiplicative complexity* can be set up via power maps.

⁷ We added the term “non-linear” for emphasizing that it is not a multiplication with a constant.

Lemma 3. Let $q = p^s$, where $p \geq 2$ is a prime and $s \geq 1$. Let $d \geq 3$ be an integer for which $x \mapsto x^d$ is invertible over $\mathbb{F}_q$, hence $\gcd(d, q-1) = 1$. Let $\alpha \in \mathbb{F}_q \setminus \{0\}$. The function

$$\mathcal{F}(x) = \sum_{i=1}^d \binom{d}{i} \cdot x^{i-1} \cdot (\pm\alpha)^{d-i} = \begin{cases} \frac{(x \pm \alpha)^d \mp \alpha^d}{x} & \text{if } x \neq 0, \\ \pm d \cdot \alpha^{d-1} & \text{otherwise} \end{cases} \quad (9)$$

satisfies the requirements of Prop. 2.

7 Cryptographic Properties of Amaryllises

In this section, we analyze the statistical and the algebraic properties of **Amaryllises**. As we are going to show, this initial analysis implies some interesting *open problems* we leave to study for future research.

Remark 2. We emphasize that the following observations do not take into account the details of the sub-components of the considered schemes. Hence, it is possible that some of the following results do *not* hold for some specific instances. Moreover, we emphasize that the following observations are based on the assumption that the following schemes are used for instantiating a MPC-/HE-/ZK-friendly symmetric primitive.

7.1 Initial Remarks: Invertibility and Full Diffusion

About the Invertibility. As we have just seen, the **Amaryllises** constructions are invertible only if their internal functions satisfy some specific conditions. Still, it is possible to turn any permutation $\mathcal{P}$ into a function $\mathcal{F}$ that satisfy them.

Computing the inverse is in general more expensive than computing the **Amaryllises** construction directly – see Eq. (8) for details. We highlight that this fact also occurs for many SPN schemes designed for MPC/HE/ZK applications and instantiated with power maps $x \mapsto x^d$ (where d is usually the smallest positive integer co-prime with $p-1$). Indeed, the inverse of $x \mapsto x^d$ is $x \mapsto x^{1/d} \equiv x^{\hat{d}}$ where $\hat{d}$ is the smallest integer for which $d \cdot \hat{d} - 1$ is a multiple of $q-1$ (due to Fermat’s Little Theorem). In the case $q \gg d$, then $\hat{d}$ is of the same order of q , making $x \mapsto x^{1/d}$ much more expensive to compute with respect to $x \mapsto x^d$. Having said that, computing the inverse of such schemes is not required in many applications, as for example:

- stream ciphers instantiated via a cipher $E_k(\cdot)$ used in a mode of operation that does not require the computation of the inverse, as the counter-mode one $(x, \mathbf{N}) \mapsto (x + E_k(\mathbf{N}), \mathbf{N})$ for a nonce $\mathbf{N}$ and a secret key k ;
- sponge hash functions [BDPA08] instantiated with permutations. In such a case, no inverse computation of the permutation is performed for computing the hash value.

Hence, the fact that computing the inverse could be very expensive does not represent a disadvantage in many practical use cases.

Regarding the Full Diffusion. Regarding the diffusion, it is trivial to check that *full diffusion* is achieved in **Amaryllises** without any additional linear layer. At the same time, an extra linear layer is crucial for destroying the invariant subspaces, as discussed in the following.

7.2 Statistical Attacks

Regarding the statistical attacks, we mainly focus on classical differential attacks and (invariant) sub-space trails (related to truncated differential attacks).

Differential (and Linear) Attacks In a differential attack [BS90, BS93], the attack exploits the probability distribution of the output differences produced by the analyzed cryptographic primitive for given input differences. Let $\delta, \Delta \in \mathbb{F}_q^n \setminus \{0\}$ be respectively the input and the output differences. We recall that the differential probability (DP) of having a certain output difference Δ given a particular input difference δ for a permutation $\mathcal{P}$ over $\mathbb{F}_q^n$ is equal to

$$\text{Prob}(\delta \neq 0 \rightarrow \Delta) = |\{x \in \mathbb{F}_q^n \mid \mathcal{P}(x + \delta) - \mathcal{P}(x) = \Delta\}|/q^n.$$

In the case of the **Amaryllises** construction, the following result holds:

Proposition 12. *The maximum DP of any Amayllises construction over $\mathbb{F}_q^n$ defined as in Theorem 2 is*

$$\leq \begin{cases} \frac{\deg(\mathcal{F})}{q} \in \mathcal{O}(q^{-1}) & \text{if } \deg(\mathcal{H}) \leq 1 \text{ (i.e., } \mathcal{H} = \text{affine function}), \\ \frac{\deg(\mathcal{F}) \cdot (\deg(\mathcal{H}) - 1)}{q^2} \in \mathcal{O}(q^{-2}) & \text{otherwise.} \end{cases}$$

Before giving the proof, we emphasize that such result is meaningful only in the case in which the degrees of the involved functions $\mathcal{F}$ and $\mathcal{H}$ are small. For example, in the case of the AES S-Box $x \mapsto x^{-1} \equiv x^{q-2}$, the result just given would be completely meaningless from a practical point of view. At the same time, the large majority of the MPC-/HE-/ZK-friendly primitives are (at least, partially) instantiated with low degree functions. As a result, in the case of an iterated primitive, the given bounds combined with the huge size of q are usually sufficient for guaranteeing security against differential cryptanalysis within few rounds.

Proof. Let $\delta \neq 0$ and Δ be respectively the input and the output differences. Each differential characteristic is defined by a system of n equations of the form

$$\begin{aligned} & x_i \cdot \left(\mathcal{F} \left(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j \right) - \mathcal{F} \left(\sum_j \beta_j \cdot x_j \right) \right) + \delta_i \cdot \mathcal{F} \left(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j \right) \\ & + \mathcal{H} \left(\sum_j \lambda_j^{(0)} \cdot (x_j + \delta_j), \sum_j \lambda_j^{(1)} \cdot (x_j + \delta_j), \dots, \sum_j \lambda_j^{(l-1)} \cdot (x_j + \delta_j) \right) \\ & - \mathcal{H} \left(\sum_j \lambda_j^{(0)} \cdot x_j, \sum_j \lambda_j^{(1)} \cdot x_j, \dots, \sum_j \lambda_j^{(l-1)} \cdot x_j \right) = \Delta_i \end{aligned} \quad (10)$$

for each $i \in \{0, 1, \dots, n-1\}$. In order to prove the result, we bound the number of possible solutions in $x_0, x_1, \dots, x_{n-1}$ of such a system of equations.

First of all, one of such equations can be replaced by their linear combination with respect to $\beta_0, \beta_1, \dots, \beta_{n-1}$. This corresponds to

$$y \cdot \left(\mathcal{F} \left(y + \sum_j \beta_j \cdot \delta_j \right) - \mathcal{F}(y) \right) + \sum_i \beta_i \cdot \delta_i \cdot \mathcal{F} \left(y + \sum_j \beta_j \cdot \delta_j \right) = \sum_i \beta_i \cdot \Delta_i$$

where $y := \sum_i \beta_i \cdot x_i$, and where either $\sum_i \beta_i = 0$ or $\mathcal{H}$ is identically equal to zero. Since such equation is of degree $\deg(\mathcal{F})$, it admits at most $\deg(\mathcal{F})$ solutions in $y = \sum_i \beta_i \cdot x_i$.

Depending on the value of $\sum_i \beta_i \cdot x_i$ just found, we examine separately the case in which $\mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j) \neq \mathcal{F}(\sum_j \beta_j \cdot x_j)$ from the case in which the equality holds.

Case: $\mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j) \neq \mathcal{F}(\sum_j \beta_j \cdot x_j)$. Let's assume that $\mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j) \neq \mathcal{F}(\sum_j \beta_j \cdot x_j)$ for a given value of $\sum_i \beta_i \cdot x_i$ found before. Then:

- assume $\mathcal{H}$ is identically equal to zero. Since $\mathcal{F}(y + \sum_j \beta_j \cdot \delta_j) \neq \mathcal{F}(y)$, for the given value of $y = \sum_i \beta_i \cdot x_i$ found before, it is possible to find the values of the remaining $n-1$ variables x_i that satisfy the system of equations in (10) by simply inverting $n-1$ equations:

$$x_i = \frac{\Delta_i - \delta_i \cdot \mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j)}{\left(\mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j) - \mathcal{F}(\sum_j \beta_j \cdot x_j) \right)}.$$

This implies that the number of solutions is $\leq \deg(\mathcal{F})$, or equivalently that the probability of the corresponding differential characteristic is $\mathcal{O}(q^{-n})$;

- if $\mathcal{H}$ is not identically equal to zero, we consider the linear combinations of the equations that compose the system of equations in (10) with respect to $\lambda_0^{(j)}, \lambda_1^{(j)}, \dots, \lambda_{n-1}^{(j)}$ for each $j \in \{0, 1, \dots, l-1\}$, that is,

$$z \cdot \left(\mathcal{F} \left(y + \sum_j \beta_j \cdot \delta_j \right) - \mathcal{F}(y) \right) + \sum_i \lambda_i^{(j)} \cdot \delta_i \cdot \mathcal{F} \left(y + \sum_j \beta_j \cdot \delta_j \right) = \sum_i \lambda_i^{(j)} \cdot \Delta_i$$

where $z := \sum_i \lambda_i^{(j)} \cdot x_i$, and where $y = \sum_j \beta_j \cdot x_j$ as before (remember that $\sum_i \lambda_i^{(j)} = 0$ by assumption). Since $\mathcal{F}\left(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j\right) \neq \mathcal{F}\left(\sum_j \beta_j \cdot x_j\right)$, it is possible to find $z = \sum_i \lambda_i^{(j)} \cdot x_i$ for each $j \in \{0, 1, \dots, l-1\}$. If $l = n-1$, this information together with the knowledge of $\sum_i \beta_i \cdot x_i$ is sufficient to recover $x_0, x_1, \dots, x_{n-1}$. If $l < n-1$, it is possible to find the remaining $n - (l+1)$ values of $x_0, x_1, \dots, x_{n-1}$ by inverting $n - (l+1)$ equations as before. In both cases, this implies that the number of solutions is $\leq \deg(\mathcal{F})$, or equivalently that the probability of the corresponding differential characteristic is $\mathcal{O}(q^{-n})$.

Case: $\mathcal{F}\left(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j\right) = \mathcal{F}\left(\sum_j \beta_j \cdot x_j\right)$. Let's assume that $\mathcal{F}(\sum_j \beta_j \cdot x_j + \sum_j \beta_j \cdot \delta_j) = \mathcal{F}(\sum_j \beta_j \cdot x_j)$ for a given value of $\sum_i \beta_i \cdot x_i$ found before. (This scenario occurs if e.g. $\sum_j \beta_j \cdot \delta_j = 0$. However, we emphasize that such equality can occur also in the case in which $\sum_j \beta_j \cdot \delta_j \neq 0$, since $\mathcal{F}$ is not bijective in general.) In such a case, by considering any difference of two equations that compose (10), note that the system of equations (10) admits a solution only if

$$\forall i, l \in \{0, 1, \dots, n-1\}, \quad (\delta_i - \delta_l) \cdot \mathcal{F}\left(\sum_j \beta_j \cdot x_j\right) = \Delta_i - \Delta_l.$$

If all these equalities are satisfied (note that they are independent of $x_0, x_1, \dots, x_{n-1}$), then for the given value of $\sum_j \beta_j \cdot x_j$ found before:

– if $\mathcal{H}$ is an affine function, that is,

$$\begin{aligned} & \mathcal{H}\left(\sum_j \lambda_j^{(0)} \cdot (x_j + \delta_j), \sum_j \lambda_j^{(1)} \cdot (x_j + \delta_j), \dots, \sum_j \lambda_j^{(l-1)} \cdot (x_j + \delta_j)\right) \\ & - \mathcal{H}\left(\sum_j \lambda_j^{(0)} \cdot x_j, \sum_j \lambda_j^{(1)} \cdot x_j, \dots, \sum_j \lambda_j^{(l-1)} \cdot x_j\right) \\ & = \mathcal{H}\left(\sum_j \lambda_j^{(0)} \cdot \delta_j, \sum_j \lambda_j^{(1)} \cdot \delta_j, \dots, \sum_j \lambda_j^{(l-1)} \cdot \delta_j\right) - \mathcal{H}(0, 0, \dots, 0), \end{aligned}$$

then no other condition on $n-1$ variables x_i is imposed. Hence, the number of solutions is $\leq \deg(\mathcal{F}) \cdot q^{n-1}$, and the probability of the corresponding differential characteristic is of order $\mathcal{O}(q^{-1})$;

– otherwise, the system of equations (10) reduces to a single equation, that is,

$$\begin{aligned} & \mathcal{H}\left(\sum_j \lambda_j^{(0)} \cdot (x_j + \delta_j), \sum_j \lambda_j^{(1)} \cdot (x_j + \delta_j), \dots, \sum_j \lambda_j^{(l-1)} \cdot (x_j + \delta_j)\right) \\ & - \mathcal{H}\left(\sum_j \lambda_j^{(0)} \cdot x_j, \sum_j \lambda_j^{(1)} \cdot x_j, \dots, \sum_j \lambda_j^{(l-1)} \cdot x_j\right) = \Delta_i, \end{aligned}$$

which admits at most $q^{n-2} \cdot (\deg(\mathcal{H}) - 1)$. It follows that the total number of solutions is $\leq \deg(\mathcal{F}) \cdot (\deg(\mathcal{H}) - 1) \cdot q^{n-2}$, and the probability of the corresponding differential characteristic is of order $\mathcal{O}(q^{-2})$.

This concludes the proof. $\square$

We point out that an analogous analysis and result can be derived for what concerning linear attacks [Mat93] as well.

Invariant Subspace Trails (and Truncated Differential) Attacks Next, we examine the existence of invariant subspace trails, recalled in Sect. 3.1. For completeness, we recall the strict link between subspace trails and truncated differentials [Knu94] pointed out in [LTW18].

In the case of **Amaryllises**, $\langle [1, 1, \dots, 1] \rangle \equiv \{[x, x, \dots, x] \mid x \in \mathbb{F}_q\} \subseteq \mathbb{F}_q^n$ is an invariant subspace for any arbitrary number of rounds. Indeed, let's consider separately the case $\sum_{i=0}^{n-1} \beta_i = 0$ from $\sum_{i=0}^{n-1} \beta_i \neq 0$:

- if $\sum_{i=0}^{n-1} \beta_i = 0$, it is simple to check that $\langle [1, 1, \dots, 1] \rangle$ forms an invariant subspace trail, since

$$\forall i \in \{0, 1, \dots, n-1\}, \quad y_i = x \cdot \underbrace{\mathcal{F}(0)}_{\neq 0} + \mathcal{H}(0, 0, \dots, 0),$$

that is, $y_i = y_j$ for each $i, j \in \{0, 1, \dots, n-1\}$ (remember that $\sum_{i=0}^{n-1} \beta_i = \sum_{i=0}^{n-1} \lambda_i^{(0)} = \sum_{i=0}^{n-1} \lambda_i^{(1)} = \dots = \sum_{i=0}^{n-1} \lambda_i^{(l-1)} = 0$ for guaranteeing the invertibility);

- if $\sum_{i=0}^{n-1} \beta_i \neq 0$ (and so $\mathcal{H}(\cdot) = 0$), we have that

$$\forall i \in \{0, 1, \dots, n-1\}, \quad y_i = x \cdot \mathcal{F}\left(x \cdot \sum_{i=0}^{n-1} \beta_i\right),$$

that is, $y_i = y_j$ for each i, j , and where note that $x \mapsto x \cdot \mathcal{F}\left(x \cdot \sum_{i=0}^{n-1} \beta_i\right)$ is a bijective function.

Indeed, since $\mathcal{G}(x) = x \cdot \mathcal{F}(x)$ is bijective, then $\frac{\mathcal{G}(x \cdot \sum_{i=0}^{n-1} \beta_i)}{\sum_{i=0}^{n-1} \beta_i} = x \cdot \mathcal{F}\left(x \cdot \sum_{i=0}^{n-1} \beta_i\right)$ is bijective as well (note that $\sum_{i=0}^{n-1} \beta_i \neq 0$).

In both cases, we point out that any invariant subspace can be broken by exploiting the results proposed in Sect. 3.1. We remark that if $l = n - 1$, then it is possible to destroy such invariant subspace in a very simple way by multiplying each y_i with a non-zero coefficient μ_i under the condition that such coefficients are pair-wise different, besides choosing proper round constants.

7.3 Algebraic Attacks

Next, we propose some considerations about the security of **Amaryllises** schemes against algebraic attacks. With respect to statistical attacks, algebraic attacks exploit the simple algebraic structure of the attacked schemes. (We recall that MPC-/HE-/ZK-friendly primitives are usually more vulnerable to algebraic attacks rather than to statistical ones.)

Growth of the Degree The simple representation of an attacked scheme mainly depends on two factors, which are the low degree and the density of the interpolation polynomials that describe the analyzed iterative primitive. Here we focus on the growth of the degree for the **Amaryllises** construction.

Regarding the forward direction, the degree of each round is obviously $\max\{1 + \deg(\mathcal{F}), \deg(\mathcal{H})\}$, while the degree of $r \geq 1$ rounds is upper bounded by $(\max\{1 + \deg(\mathcal{F}), \deg(\mathcal{H})\})^r$. Without considering the details of the involved functions, it is hard to derive better bounds. For this reason, here we mainly focus on the backward direction.

Referring to the proof of Theorem 2, we highlight that, while $x \mapsto x \cdot \mathcal{F}(x)$ is evaluated in the forward direction, the function $x \mapsto \mathcal{F}(\mathcal{G}^{-1}(x))$ is evaluated in the backward direction (where $\mathcal{G}(x) := x \cdot \mathcal{F}(x)$). The crucial point we aim to emphasize is that the degree of $\mathcal{F} \circ \mathcal{G}^{-1}$ could be much higher than the degree of $\mathcal{F}$. As a concrete example, consider the case in which $\mathcal{F}$ is defined via an invertible power map as in (9), where $d \geq 3$ is the smallest integer co-prime with $q - 1$. (As recalled before, $x \mapsto x^{1/d} \equiv x^{\hat{d}}$ where $\hat{d}$ is the smallest integer for which $d \cdot \hat{d} - 1$ is a multiple of $q - 1$. If $q \gg d$, then $\hat{d}$ is of the same order of q .) In such a case, we have that

$$\mathcal{F} \circ \mathcal{G}^{-1}(x) := \begin{cases} \left(((x \pm \alpha^d)^{1/d} \mp \alpha)^d \pm \alpha^d \right) \cdot ((x \pm \alpha^d)^{1/d} \mp \alpha)^{-1} & \text{if } x \neq 0, \\ \pm d \cdot \alpha^{d-1} & \text{otherwise} \end{cases}$$

which implies that $\deg(\mathcal{F} \circ \mathcal{G}^{-1})$ is close to the maximum (hence, close to q), that is, much higher than the degree $d - 1$ of $\mathcal{F}$.

This fact has a crucial impact in the security against e.g. Meet-in-the-Middle (MitM) algebraic attacks. In such a case, it is crucial to reach the maximum degree (or a sufficiently high degree depending on the security level) both in the forward and in the backward direction. *The Amaryllises constructions guarantee that the maximum degree can be reached within few rounds in the backward direction even if the degree of $\mathcal{F}$ is small.* This implies concrete benefits for what concern the multiplicative complexity of **Amaryllises**, since fewer rounds are potentially required for achieving security with respect to other non-linear schemes/networks/constructions proposed in the literature.

Specific Algebraic Attacks: an Open Problem related to Gröbner Basis As pointed out before, the degree is not the only ingredient that influences the cost of an algebraic attack. Other factors such as the density of the interpolation polynomials, the number of equations and variables, among others, play a crucial role as well. However, all these factors strictly depend on the details of the functions that instantiate each scheme. Hence, making claim about the security against generic algebraic attacks for generic constructions is quite hard (and probably meaningless).

For this reason, we limit ourselves to point out an interesting open problem for future work regarding the Gröbner basis attack [Buc76]. Gröbner basis is a strategy that allows to find solution(s) – if they exist – of a given system of non-linear equations that describe the analyzed scheme (depending on the scheme, the variable could be either the key for a cipher or a pre-image/collision for an hash function). In [GHR⁺23, Sect. 6.3], GRIFFIN’s designers noticed that the **Horst** construction provides concrete advantages in defeating the Gröbner basis attack with respect to a Feistel scheme due to the non-linear combination between x_i and the function $\mathcal{G}_i(x_0, \dots, x_{i-1})$. As a result, even if a single round of **Horst** is clearly more expensive than a Feistel round (from the multiplicative point of view), the **Horst** construction has concrete advantages both in terms of security and performances over multiple rounds. Still, *a formal theoretical argument that supports this observation for Horst is still missing*, and open for future research.

A similar result/scenario can occur for **Amaryllises** as well. Based on that, we leave the following *open problem for future work*:

does the non-linear mixing in **Amaryllises** provide concrete advantages similar to the case of **Horst** with respect to SPN (or other) schemes in the case of Gröbner basis attacks?

7.4 Algebraic Properties: Analogies and Advantages with respect to other Cryptographic Primitives

Focusing on the case in which **Amaryllises** is used in MPC/HE/ZK scenarios (see Remark 2), here we would like to compare the algebraic properties of **Amaryllises** with respect to other cryptographic primitives published in the literature. Having said of the open problem related to the Gröbner basis attacks (see previous section), here we focus on the growth of the degree.

SPN. Regarding the growth of the degree, the conclusions given for **Amaryllises** hold for SPN schemes as well. Indeed, while the S-Box is computed in the forward direction, its inverse is computed in the backward direction. If the inverse S-Box has a degree that is much higher than the degree of the S-Box itself (as for the case of $x \mapsto x^d$ versus $x \mapsto x^{1/d}$), few rounds are required to reach the maximum degree (or a sufficiently high degree) in the backward direction. (We limit ourselves to point out that the degree can grow very fast in both direction for particular choices of the S-Boxes, such as $x \mapsto 1/x$. However, this is not often the case for recent MPC-/HE-/ZK-friendly schemes).

Feistel and Lai-Massey Schemes. Regarding Feistel networks and Lai-Massey schemes, several scenarios are possible. In some cases, the growth of the degree is the same in both the forward and the backward direction, and others in which the degree grows faster in the backward direction than in the forward one. As concrete examples, in a Type-II Feistel network $[x_0, x_1, \dots, x_{2n-1}] \mapsto [y_0, y_1, \dots, y_{2n-1}]$, the growth of the degree is equal in the two directions, since for each $i \in \{0, 1, \dots, 2n-1\}$:

$$y_i := \begin{cases} x_{i+1} + \mathcal{F}(x_i) & \text{if } i \bmod 2 = 0, \\ x_{i+1} & \text{otherwise,} \end{cases} \quad \text{versus} \quad x_i = \begin{cases} y_{i-1} & \text{if } i \bmod 2 = 0, \\ y_{i-1} - \mathcal{F}(y_{i-2}) & \text{otherwise.} \end{cases}$$

A similar conclusion holds for e.g. the Lai-Massey scheme defined in Prop. 1. In all these cases, the same number of rounds is necessary to reach the same (maximum) degree in both the forward and the backward direction.

For comparison, given a Type-III Feistel network defined as

$$[x_0, \dots, x_{n-2}, x_{n-1}] \mapsto [y_0, \dots, y_{n-2}, y_{n-1}] := [x_1 + \mathcal{F}(x_0), \dots, x_{n-1} + \mathcal{F}(x_{n-2}), x_0],$$

its inverse is given by

$$x_0 = y_{n-1}, \quad x_1 = y_0 - \mathcal{F}(y_{n-1}), \quad x_2 = y_1 - \mathcal{F}(y_0 - \mathcal{F}(y_{n-1})), \quad \dots$$

In such a case, the degree of the inverse in the i -th $\mathbb{F}_q$ -word is upper bounded by $(\deg(\mathcal{F}))^{i-1} \leq (\deg(\mathcal{F}))^{n-1}$, which is higher than the degree of the corresponding forward function (that is, $\deg(\mathcal{F})$). A similar result/conclusion holds for e.g. the generalized Lai-Massey schemes GLM_n proposed in Prop. 7 – 8. In these cases, a smaller number of rounds is potentially sufficient in order to achieve the maximum degree (or a sufficiently high degree – see before) in the backward direction with respect to the one necessarily in the forward direction.

Horst Construction. Regarding the **Horst** construction, the main difference between the forward and the backward direction relies on the fact that a division takes places instead of a multiplication:

$$y_i = x_i \cdot \mathcal{G}_i(x_0, \dots, x_{i-1}) + \mathcal{F}_i(x_0, \dots, x_{i-1}) \quad \text{versus} \quad x_i = \frac{y_i - \mathcal{F}_i(x_0, \dots, x_{i-1})}{\mathcal{G}_i(x_0, \dots, x_{i-1})},$$

for given $x_0, x_1, \dots, x_{i-1}$ (assuming $i \geq 1$). In such a case, the growth of the degree in the backward direction depends both on the “representation” and on the details of the functions $\mathcal{G}_0, \mathcal{G}_1, \dots, \mathcal{G}_{n-2}$. Regarding the representation, the attacker can describe the scheme via a fraction of polynomials, as originally proposed by Jakobsen and Knudsen in the interpolation attack against modified versions of SHARK instantiated with $x \mapsto 1/x$ (see [JK97, Sect. 3.4] for more details). This could help in keeping the degree of the denominator and of the nominator low, since the overall degree of each fraction is at most $\max\{1, \deg(\mathcal{F}_i), \deg(\mathcal{G}_i)\}$. Yet, working over multiple rounds may require to combine the fractions, and this fact can impact on the growth of the degree over multiple rounds:

- in the extreme case in which all functions $\mathcal{G}_i$ are equal, then combining the fractions does not increase the degree of the common denominator;
- otherwise, one has to compute the least common multiple $\text{lcm}(\mathcal{G}_0, \mathcal{G}_1, \dots, \mathcal{G}_{n-2})$, which could be of degree much higher than the single denominators $\mathcal{G}_0, \mathcal{G}_1, \dots, \mathcal{G}_{n-2}$.

Finally, if it is not possible to work with the fraction representation, then the degree of the polynomial corresponding to $1/\mathcal{G}_i$ could be very large (as a simple example, the degree of $x \mapsto 1/x \equiv x^{p-2}$ over $\mathbb{F}_p$ is $p-2$).

8 The Contracting-Amaryllises Construction

In this section, we introduce the contracting-**Amaryllises** constructions. With respect to the construction proposed in Theorem 2, here the function $\mathcal{F}$ is defined as $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$, that is, it takes n $\mathbb{F}_q$ -elements as input and returns a single $\mathbb{F}_q$ -element, as the name “contracting” suggests.

Proposition 13 (Contracting-Amaryllises). *Let $q = p^s$ be as before, and let $n \geq 2$ be an integer. Let $e \geq 1$ be an integer such that $\gcd(e, q-1) = 1$. Let $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ be a function that never returns zero for any non-zero input, that is,*

$$\forall [x_0, x_1, \dots, x_{n-1}] \in \mathbb{F}_q^n \setminus \{[0, 0, \dots, 0]\}, \quad \mathcal{F}(x_0, x_1, \dots, x_{n-1}) \neq 0.$$

If the function $\mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}(x) : \mathbb{F}_q \rightarrow \mathbb{F}_q$ defined as

$$\mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}(x) := x^e \cdot \mathcal{F}(\psi_0 \cdot x, \psi_1 \cdot x, \dots, \psi_{n-1} \cdot x)$$

is invertible for each arbitrary fixed non-null $[\psi_0, \psi_1, \dots, \psi_{n-1}] \in \mathbb{F}_q^n \setminus \{[0, 0, \dots, 0]\}$, then the contracting-Amaryllises construction A_C over $\mathbb{F}_q^n$ defined as $A_C(x_0, x_1, \dots, x_{n-1}) = y_0 \|y_1\| \dots \|y_{n-1}$ where

$$\forall i \in \{0, 1, \dots, n-1\}, \quad y_i := x_i^e \cdot \mathcal{F}(x_0, x_1, \dots, x_{n-1}) \quad (11)$$

is invertible.

Proof. We start by pointing out two observations:

- first of all, the following equality always holds:

$$\forall i, j \in \{0, 1, \dots, n-1\}, \quad y_i \cdot x_j^e = y_j \cdot x_i^e = x_i^e \cdot x_j^e \cdot \mathcal{F}(x_0, x_1, \dots, x_{n-1}); \quad (12)$$

- secondly, $x_i = 0$ if and only if $y_i = 0$.

Regarding this second point, note that if $x_i = 0$, then $y_i = 0$. Vice-versa, if $y_i = 0$, then either $x_i^e = 0$ (and so $x_i = 0$) or $\mathcal{F}(x_0, x_1, \dots, x_{n-1}) = 0$. However, $\mathcal{F}(x_0, x_1, \dots, x_{n-1}) = 0$ if and only if $[x_0, x_1, \dots, x_{n-1}] = [0, 0, \dots, 0]$, which implies again $x_i = 0$.

For each $i \in \{0, 1, \dots, n-1\}$ such that $y_i \neq 0$ (remember that $y_i = 0$ implies $x_i = 0$):

$$\begin{aligned} y_i &= x_i^e \cdot \mathcal{F} \left(\left(\frac{y_0}{y_i} \right)^{\frac{1}{e}} \cdot x_i, \dots, \left(\frac{y_{i-1}}{y_i} \right)^{\frac{1}{e}} \cdot x_i, x_i, \left(\frac{y_{i+1}}{y_i} \right)^{\frac{1}{e}} \cdot x_i, \dots, \left(\frac{y_{n-1}}{y_i} \right)^{\frac{1}{e}} \cdot x_i \right) \\ &= \mathcal{G}_{\left(\frac{y_0}{y_i} \right)^{\frac{1}{e}}, \dots, \left(\frac{y_{i-1}}{y_i} \right)^{\frac{1}{e}}, 1, \left(\frac{y_{i+1}}{y_i} \right)^{\frac{1}{e}}, \dots, \left(\frac{y_{n-1}}{y_i} \right)^{\frac{1}{e}}} (x_i), \end{aligned}$$

due to (12), and where $x \mapsto x^e$ is invertible by assumption on e . Since $\mathcal{G}$ is invertible by assumption (note that $\psi_j = (y_j/y_i)^{1/e}$ is fixed for each $j \in \{0, 1, \dots, n-1\}$), it is always possible to recover x_i for each $i \in \{0, 1, \dots, n-1\}$ as

$$x_i = \begin{cases} 0 & \text{if } y_i = 0, \\ \mathcal{G}_{\left(\frac{y_0}{y_i} \right)^{\frac{1}{e}}, \dots, \left(\frac{y_{i-1}}{y_i} \right)^{\frac{1}{e}}, 1, \left(\frac{y_{i+1}}{y_i} \right)^{\frac{1}{e}}, \dots, \left(\frac{y_{n-1}}{y_i} \right)^{\frac{1}{e}}}^{-1}(y_i) & \text{otherwise.} \end{cases}$$

□

The contracting-Amayllises construction does *not* admit invariant subspaces in general, since the inputs of the function $\mathcal{F}$ are $x_0, x_1, \dots, x_{n-1}$ directly, and not linear combinations of them. However, such invariant subspaces can exist depending on the details of the function $\mathcal{F}$ itself.

8.1 Homogeneous Functions for the contracting-Amayllises Construction

The challenge we now have to face regards the construction of functions $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ that (i) satisfy the assumptions of Prop. 13, and that (ii) are efficient to compute from the multiplicative point of view.

In Prop. 14, we prove that homogeneous functions that never return zero satisfy such assumptions. Based on it, in the next subsection, we then provide some concrete examples of such functions.

Proposition 14. *Let $q = p^s$ be as before, and let $n \geq 1$. Let $d \geq 3$ be such that $\gcd(d, q-1) = 1$, and let $e \geq 1$ be an integer such that (i) $\gcd(e, q-1) = 1$ and such that (ii) $d' := d - e \geq 0$. Let $\mathcal{J}_{d'} := \left\{ [i_0, i_1, \dots, i_{n-1}] \in \mathbb{Z}_+^n \mid \sum_{j=0}^{n-1} i_j = d' \right\}$.*

A function $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ satisfies the assumptions of Prop. 13 if the following conditions are fulfilled:

1. *$\mathcal{F}$ is a homogeneous function of degree d' (that is, a sum of monomials of degree d' only) as*

$$\mathcal{F}(x_0, x_1, \dots, x_{n-1}) = \sum_{\{i_0, i_1, \dots, i_{n-1}\} \in \mathcal{J}_{d'}} \varphi_{i_0, i_1, \dots, i_{n-1}} \cdot x_0^{i_0} \cdot x_1^{i_1} \cdot \dots \cdot x_{n-1}^{i_{n-1}},$$

where $\varphi_{i_0, i_1, \dots, i_{n-1}} \in \mathbb{F}_q$;

2. *$\mathcal{F}$ never returns zero for any non-zero input, that is, $\mathcal{F}(x_0, x_1, \dots, x_{n-1}) \neq 0$ for each $[x_0, x_1, \dots, x_{n-1}] \in \mathbb{F}_q^n \setminus \{[0, 0, \dots, 0]\}$.*

Proof. It is sufficient to prove that $\mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}(x) = x^{d-d'} \cdot \mathcal{F}(\psi_0 \cdot x, \psi_1 \cdot x, \dots, \psi_{n-1} \cdot x)$ is invertible for each arbitrary fixed non-null $[\psi_0, \psi_1, \dots, \psi_{n-1}] \in \mathbb{F}_q^n \setminus \{[0, 0, \dots, 0]\}$.

Since $\mathcal{F}$ contains only monomials of degree d' , then

$$\mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}(x) = x^e \cdot \mathcal{F}(\psi_0 \cdot x, \psi_1 \cdot x, \dots, \psi_{n-1} \cdot x) = x^d \cdot \mathcal{F}(\psi_0, \psi_1, \dots, \psi_{n-1}),$$

where $d = d' + e$ by definition, and since $\mathcal{F}$ is homogeneous of degree d . Since (i) $x \mapsto x^d$ is invertible due to the assumption on d and since (ii) $\mathcal{F}$ never returns zero for each non-null input by assumption, then the inverse of $y = \mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}(x)$ is given by

$$x = \mathcal{G}_{\psi_0, \psi_1, \dots, \psi_{n-1}}^{-1}(y) = \left(\frac{y}{\mathcal{F}(\psi_0, \psi_1, \dots, \psi_{n-1})} \right)^{\frac{1}{d}}.$$

This concludes the proof. □

8.2 Suitable Functions for the contracting-Amayllises Construction and Open Problems

Suitable Functions $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ Here, we start by proposing some concrete examples of functions from $\mathbb{F}_q^2$ into $\mathbb{F}_q$ that satisfy the conditions just given in Prop. 14.

Lemma 4. *Given $q = p^s$ as before, let $d \geq 3$ be such that $\gcd(d, q-1) = 1$, and let $d' = d-1$ (equivalently, $e = 1$). Let $\alpha, \beta \in \mathbb{F}_q \setminus \{0\}$. The function $\mathcal{F} : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ defined as*

$$\mathcal{F}(x_0, x_1) = \sum_{i=1}^d \binom{d}{i} \cdot \alpha^i \cdot \beta^{d-i} \cdot x_0^{i-1} \cdot x_1^{d-i} = \begin{cases} \frac{(\alpha \cdot x_0 + \beta \cdot x_1)^d - (\beta \cdot x_1)^d}{x_0} & \text{if } x_0 \neq 0, \\ d \cdot \alpha \cdot \beta^{d-1} \cdot x_1^{d-1} & \text{otherwise,} \end{cases}$$

satisfies the conditions given in Prop. 14.

Proof. The proof is trivial. Indeed, it is obvious that the function $\mathcal{F}$ is homogeneous of degree $d' = d-1$. Moreover, it never returns zero for any non-zero input, since (i) $d \cdot \alpha \cdot \beta^{d-1} \cdot x_1^{d-1} = 0$ if and only if $x_0 = x_1 = 0$, and (ii) $\frac{(\alpha \cdot x_0 + \beta \cdot x_1)^d - (\beta \cdot x_1)^d}{x_0} = 0$ if and only if $(\alpha \cdot x_0 + \beta \cdot x_1)^d = (\beta \cdot x_1)^d$, that is, $x_0 = 0$, which is not possible by assumption. $\square$

An example for the prime fields only is proposed in the following.

Lemma 5. *Let $p \geq 3$ be a prime integer, and let $d \geq 3$ be such that $\gcd(d, p-1) = 1$. Let $d' \in \{2, 4, \dots, d-1\}$ be an even integer smaller than d such that $\gcd(d-d', p-1) = 1$. Let $\alpha, \beta, \lambda, \lambda', \omega \in \mathbb{F}_p$ be such that (i) $\lambda \neq \lambda'$ and (ii) ω is a quadratic non-residue modulo p . The function*

$$\mathcal{F}(x_0, x_1) = \alpha^2 \cdot (x_0 + \lambda \cdot x_1)^{d'} - \omega \cdot \beta^2 \cdot (x_0 + \lambda' \cdot x_1)^{d'}$$

satisfies the assumptions of Prop. 14.

Proof. It is sufficient to show that $\mathcal{F}(x_0, x_1) \neq 0$ for each $[x_0, x_1] \neq [0, 0]$. Assume by contradiction that there exists $[x_0, x_1] \neq [0, 0]$ such that $\mathcal{F}(x_0, x_1) = 0$, that is, $\left(\alpha \cdot (x_0 + \lambda \cdot x_1)^{\frac{d'}{2}}\right)^2 = \omega \cdot \left(\beta \cdot (x_0 + \lambda' \cdot x_1)^{\frac{d'}{2}}\right)^2$. Such equality is satisfied only in the case where both sides are equal to zero, since the left-hand side of the equality is a quadratic residue modulo p , while the right-hand side is a quadratic non-residue modulo p , due to the choice of ω . However, note that $x_0 + \lambda \cdot x_1 = x_0 + \lambda' \cdot x_1 = 0$ occurs if and only if $x_0 = x_1 = 0$, since the vectors $[1, \lambda] \in \mathbb{F}_p^2$ and $[1, \lambda'] \in \mathbb{F}_p^2$ are linearly independent (since $\lambda \neq \lambda'$). Hence, if $x_0 \neq 0$ or/and $x_1 \neq 0$, such equality never holds. $\square$

Suitable Functions $\mathbb{F}_q^{\geq 3} \rightarrow \mathbb{F}_q$ Next, we generalize the previous results for the case $\mathbb{F}_q^n \rightarrow \mathbb{F}_q$ with $n \geq 3$. Our strategy is to construct the functions $\mathcal{F}$ that satisfy Prop. 14 in an iterated way, that is, given a function $\mathcal{F}_m : \mathbb{F}_q^m \rightarrow \mathbb{F}_q$ for a certain $m \geq 2$ that satisfies the required properties, we show how to construct a function $\mathcal{F}_n : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ for $n > m$ that satisfies the required properties as well.

Proposition 15. *Given $q = p^s$ as before, let $m \geq 2$ and let $n_0, n_1, \dots, n_{m-1} \geq 1$ and let $n := \sum_{i=0}^{m-1} n_i$. For each $i \in \{n_0, n_1, \dots, n_{m-1}, m\}$, let $\mathcal{F}_i : \mathbb{F}_q^{n_i} \rightarrow \mathbb{F}_q$ be a function that satisfy the assumptions of Prop. 14, that is, (i) it is an homogeneous function of a certain degree $\deg(\mathcal{F}_i) \geq 1$, and (ii) it never returns zero for any non-zero input (i.e., $\mathcal{F}_i(x_0, x_1, \dots, x_{i-1}) \neq 0$ for each $[x_0, x_1, \dots, x_{i-1}] \in \mathbb{F}_q^i \setminus \{[0, 0, \dots, 0]\}$).*

Let $d \geq 2$ be the least common multiple of $\deg(\mathcal{F}_{n_0}), \deg(\mathcal{F}_{n_1}), \dots, \deg(\mathcal{F}_{n_{m-1}})$, that is,

$$d := \text{lcm}(\deg(\mathcal{F}_{n_0}), \deg(\mathcal{F}_{n_1}), \dots, \deg(\mathcal{F}_{n_{m-1}})) .$$

The function $\mathcal{F}_n : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ defined as

$$\mathcal{F}_n(x_0, x_1, \dots, x_n) := \mathcal{F}_m \left((\mathcal{F}_{n_0}(x_0, \dots, x_{n_0-1}))^{\frac{d}{\deg(\mathcal{F}_{n_0})}}, (\mathcal{F}_{n_1}(x_{n_0}, \dots, x_{n_0+n_1-1}))^{\frac{d}{\deg(\mathcal{F}_{n_1})}}, \dots, (\mathcal{F}_{n_{m-1}}(x_{n-n_m}, \dots, x_{n-1}))^{\frac{d}{\deg(\mathcal{F}_{n_{m-1}})}} \right)$$

satisfies the assumptions of Prop. 14, that is,

1. it is homogeneous of degree $d \cdot \deg(\mathcal{F}_m)$;
2. $\mathcal{F}_n$ never returns zero for any non-zero input in $\mathbb{F}_q^n$.

Proof. Regarding the first point, $\mathcal{F}_n$ is a homogeneous function of degree $d \cdot \deg(\mathcal{F}_m)$ since (i) $\mathcal{F}_m$ is a homogeneous function of degree $\deg(\mathcal{F}_m)$ and (ii) each input of $\mathcal{F}_m$ is a homogeneous function of degree d .

Regarding the second point, $\mathcal{F}_n$ returns zero if and only if all its inputs are equal to zero since (i) $\mathcal{F}_m$ returns zero if and only if all its inputs are equal to zero and (ii) each input of $\mathcal{F}_m$, that is, $\mathcal{F}_{n_i}(z_0, z_1, \dots, z_{n_i-1})$, returns zero if and only if $z_0 = z_1 = \dots = z_{n_i-1} = 0$. $\square$

By applying the previous result iteratively, it is possible to construct functions $\mathcal{F}_n : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ that satisfy the assumptions of Prop. 14 for each $n \geq 3$ as

$$\begin{aligned} \mathcal{F}_n(x_0, x_1, \dots, x_{n-1}) &= \mathcal{F}_2\left(\mathcal{F}_{n-1}(x_0, x_1, \dots, x_{n-2}), x_{n-1}^{\deg(\mathcal{F}_{n-1})}\right), & \text{or} \\ \mathcal{F}_{2n}(x_0, x_1, \dots, x_{2n-1}) &= \mathcal{F}_2\left(\mathcal{F}_n(x_0, x_1, \dots, x_{n-1}), \mathcal{F}_n(x_n, x_{n+1}, \dots, x_{2n-1})\right), \end{aligned}$$

by making use of the functions $\mathcal{F}_2 : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ proposed before.

Open Problem. The main drawback of this strategy regards the fact that the degrees of the obtained functions are *strictly* bigger than the degrees of the input functions. We leave the problem to propose low-degree functions $\mathcal{F}_n$ that satisfy the required assumptions of Prop. 14 as an open problem for future work.

9 Summary and Future Directions

In this paper, we re-considered the Lai-Massey scheme originally proposed in [LM90, Vau99], and we presented new generalizations that are not (extended) affine equivalent to any generalized Feistel network. Inspired by the recent **Horst** construction, we also introduced the **Amaryllises** construction, in which the linear combination that takes place in the Lai-Massey scheme can be replaced by a non-linear one. An initial analysis of its statistical and algebraic properties is provided, showing its possible advantages when used in the context of MPC-/HE-/ZK-friendly primitives.

The initial results proposed in this paper may open up *new interesting scenarios regarding the construction of new non-linear layers for future MPC-/HE-/ZK-friendly designs*. For this reason, we propose some open problems that could be interesting to analyze for future works:

- given the variants of the Lai-Massey schemes proposed in this paper that are not EA-equivalent to any Feistel network, is it possible to identify the ones with better statistical properties than the Feistel networks (for a similar cost)?
- check if the CCZ equivalence [CCZ98, CP19] holds or not among (some of) the networks/schemes presented in this paper;
- propose new generalizations of the Lai-Massey schemes and of the **Amaryllises** constructions, and/or propose new concrete efficient instantiations of the constructions presented in this paper. As a concrete example, in Sect. 8, we proposed a variant of the **Amaryllises** construction called **Contracting-Amaryllises**. At the current state, it is not clear how to efficiently instantiate it;
- better understand the advantages and the disadvantages of the **Horst** and of the **Amaryllises** constructions when used to instantiate a MPC-/HE-/ZK-friendly primitive, with particular attention to the case of Gröbner Basis attacks.

Acknowledgments. The author thanks the FSE/ToSC’22 Reviewers for their valuable suggestions and comments. Lorenzo Grassi was supported by the European Research Council (ERC), grant number 101160608 “SYMPZON”.

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A Details – Proof of Prop. 4 for $n \geq 3$

A.1 About Lemma 6

Assume that $\lambda_i^{(j)} \in \mathbb{F}_q$ satisfy the following conditions:

- i. $\sum_{j=0}^{n-1} \lambda_j^{(i)} = 0$ for each $i \in \{0, 1, \dots, n-2\}$;
- ii. the vectors $\bar{\lambda}^{(0)} = [\lambda_0^{(0)}, \lambda_1^{(0)}, \dots, \lambda_{n-1}^{(0)}]$, $\bar{\lambda}^{(1)} = [\lambda_0^{(1)}, \lambda_1^{(1)}, \dots, \lambda_{n-1}^{(1)}]$, $\dots$, $\bar{\lambda}^{(n-2)} = [\lambda_0^{(n-2)}, \lambda_1^{(n-2)}, \dots, \lambda_{n-1}^{(n-2)}] \in \mathbb{F}_q^n \setminus \{0\}$ are linearly independent.

Then, the following result holds.

Lemma 6. *Given q and n as before, let $\bar{\lambda}^{(0)}, \bar{\lambda}^{(1)}, \dots, \bar{\lambda}^{(n-2)} \in \mathbb{F}_q^n$ be $n-1$ vectors that satisfy the previous two conditions just given. Then, the vectors $\hat{\lambda}^{(0)} = [\lambda_0^{(0)}, \lambda_1^{(0)}, \dots, \lambda_{n-2}^{(0)}]$, $\hat{\lambda}^{(1)} = [\lambda_0^{(1)}, \lambda_1^{(1)}, \dots, \lambda_{n-2}^{(1)}]$, $\dots$, $\hat{\lambda}^{(n-2)} = [\lambda_0^{(n-2)}, \lambda_1^{(n-2)}, \dots, \lambda_{n-2}^{(n-2)}] \in \mathbb{F}_q^{n-1}$ (i.e., the previous vectors without the final component) are linearly independent as well.*

Proof. Assume by contradiction that there exist (non-trivial) $\psi_0, \psi_1, \dots, \psi_{n-2} \in \mathbb{F}_q$ such that $\sum_{j=0}^{n-2} \psi_j \cdot \hat{\lambda}^{(j)} = 0 \in \mathbb{F}_q^{n-1}$. This also implies that $\sum_{j=0}^{n-2} \psi_j \cdot \bar{\lambda}^{(j)} = 0 \in \mathbb{F}_q^n$ as well, since

- for each $i \in \{0, 1, \dots, n-2\}$, we have that $\sum_{j=0}^{n-2} \psi_j \cdot \lambda_i^{(j)} = 0 \in \mathbb{F}_q$, due to the fact that $\sum_{j=0}^{n-2} \psi_j \cdot \hat{\lambda}^{(j)} = 0 \in \mathbb{F}_q^{n-1}$;
- about the last component,

$$\sum_{j=0}^{n-2} \psi_j \cdot \lambda_{n-1}^{(j)} = \sum_{j=0}^{n-2} \psi_j \cdot \left(- \sum_{i=0}^{n-2} \lambda_i^{(j)} \right) = - \sum_{i=0}^{n-2} \left(\sum_{j=0}^{n-2} \psi_j \cdot \lambda_i^{(j)} \right) = \sum_{i=0}^{n-2} 0 = 0 \in \mathbb{F}_q,$$

where the first equality follows from $\sum_{j=0}^{n-1} \lambda_j^{(i)} = 0 \in \mathbb{F}_q$ for each $i \in \{0, 1, \dots, n-2\}$, while the third one follows from $\sum_{j=0}^{n-2} \psi_j \cdot \hat{\lambda}^{(j)} = 0 \in \mathbb{F}_q^{n-1}$.

This contradicts the second condition of linear independence among $\bar{\lambda}^{(0)}, \bar{\lambda}^{(1)}, \dots, \bar{\lambda}^{(n-2)}$. $\square$

A.2 Details about $A \times (B \times \text{circ}(0, 1, 0, \dots, 0)) = I$

Here we show that

$$A \times (B \times \text{circ}(0, 1, 0, \dots, 0)) = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ \lambda_0^{(0)} & \lambda_1^{(0)} & \lambda_2^{(0)} & \dots & \lambda_{n-1}^{(0)} \\ \lambda_0^{(1)} & \lambda_1^{(1)} & \lambda_2^{(1)} & \dots & \lambda_{n-1}^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{(n-2)} & \lambda_1^{(n-2)} & \lambda_2^{(n-2)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & \mu_{1,0} & \mu_{1,1} & \dots & \mu_{1,n-2} \\ 1 & \mu_{2,0} & \mu_{2,1} & \dots & \mu_{2,n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \mu_{n-1,0} & \mu_{n-1,1} & \dots & \mu_{n-1,n-2} \end{bmatrix} = I$$

is again the identity matrix. Indeed, by re-writing Eq. (5), we get

$$\begin{bmatrix} \lambda_0^{(0)} & \lambda_0^{(1)} & \dots & \lambda_0^{(n-2)} \\ \lambda_1^{(0)} & \lambda_1^{(1)} & \dots & \lambda_1^{(n-2)} \\ \lambda_2^{(0)} & \lambda_2^{(1)} & \dots & \lambda_2^{(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{n-1}^{(0)} & \lambda_{n-1}^{(1)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix} \times \begin{bmatrix} \mu_{1,0} & \mu_{2,0} & \dots & \mu_{n-1,0} \\ \mu_{1,1} & \mu_{2,1} & & \mu_{n-1,1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{1,n-2} & \mu_{2,n-2} & & \mu_{n-1,n-2} \end{bmatrix} = \begin{bmatrix} -1 & -1 & \dots & -1 \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix},$$

that is,

$$\underbrace{\begin{bmatrix} \lambda_1^{(0)} & \lambda_1^{(1)} & \dots & \lambda_1^{(n-2)} \\ \lambda_2^{(0)} & \lambda_2^{(1)} & \dots & \lambda_2^{(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{n-1}^{(0)} & \lambda_{n-1}^{(1)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix}}_{\equiv \hat{A}} \times \underbrace{\begin{bmatrix} \mu_{1,0} & \mu_{2,0} & \dots & \mu_{n-1,0} \\ \mu_{1,1} & \mu_{2,1} & & \mu_{n-1,1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{1,n-2} & \mu_{2,n-2} & & \mu_{n-1,n-2} \end{bmatrix}}_{\equiv \hat{B}} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}.$$

Hence, given $\hat{A}, \hat{B} \in \mathbb{F}_q^{(t-1) \times (t-1)}$ such that $\hat{A} \times \hat{B} = I$, we also have that $\hat{B} \times \hat{A} = I$ and that $(\hat{B} \times \hat{A})^T = \hat{A}^T \times \hat{B}^T = I^T = I$, that is,

$$\begin{bmatrix} \lambda_1^{(0)} & \lambda_2^{(0)} & \dots & \lambda_{n-1}^{(0)} \\ \lambda_1^{(1)} & \lambda_2^{(1)} & \dots & \lambda_{n-1}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{(n-2)} & \lambda_2^{(n-2)} & \dots & \lambda_{n-1}^{(n-2)} \end{bmatrix} \times \begin{bmatrix} \mu_{1,0} & \mu_{1,1} & \dots & \mu_{1,n-2} \\ \mu_{2,0} & \mu_{2,1} & \dots & \mu_{2,n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{n-1,0} & \mu_{n-1,1} & \dots & \mu_{n-1,n-2} \end{bmatrix} = I.$$

The result $A \times (B \times \text{circ}(0, 1, 0, \dots, 0)) = I$ follows immediately.⁸

B Proof of Prop. 8

Here, we prove that the construction proposed in Prop. 8 is invertible.

The invertibility is proven by working iteratively, keeping in mind that GLM_4 is invertible (see Prop. 7 for details). Let's assume that GLM_{n-2} is invertible. It follows immediately that it is possible to recover $x_0 - x_1, x_1 - x_2, \dots, x_{n-4} - x_{n-3}$ by $y_0, y_1, \dots, y_{n-3}$, due to the fact that such differences are independent of the last two outputs. Indeed, by construction, for each $i \in \{0, 1, \dots, n-3\}$, the output y_i depends only on $w_0, w_1, \dots, w_{n-4}$ and on $\mathcal{F}_{n-1}(w_0, w_1, \dots, w_{n-4}, w_{n-3}, w_{n-2})$ in such a way that the difference $y_i - y_j$ is independent of $\mathcal{F}_{n-1}(w_0, w_1, \dots, w_{n-4}, w_{n-3}, w_{n-2})$.

⁸ This also implies that both A and B are invertible, since $\det(A \times (B \times \text{circ}(0, 1, 0, \dots, 0))) = \det(I) = 1$ implies that $\det(A) \cdot \det(B) \neq 0$, and so $\det(A), \det(B) \neq 0$.

Next, given $w_0 = x_0 - x_1, w_1 = x_1 - x_2, \dots, w_{n-4} = x_{n-4} - x_{n-3}$, we have to find $w_{n-3} = x_{n-3} - x_{n-2}$ and $w_{n-2} = x_{n-2} - x_{n-1}$ in order to invert the system. By simple computation:

$$x_{n-2} - x_{n-1} = y_{n-2} - y_{n-1},$$

$$x_{n-3} - x_{n-2} = y_{n-3} - y_{n-2} - \sum_{i=1}^{n-3} \mathcal{F}_i(w_0, w_1, \dots, w_{i-1}) - \mathcal{F}_{n-2}(w_0, w_1, \dots, w_{n-4}, w_{n-2}),$$

where the r.h.s. of this last equation is independent of w_{n-3} by construction. Working exactly as before, given w_i for each $i \in \{0, 1, \dots, n-2\}$, it is possible to invert the system and recover $x_0, x_1, \dots, x_{n-1}$. This concludes the proof. $\square$